THE AI LITERACY DIVIDE: ACCESS, CAPABILITY, AND INEQUALITY IN THE AGE OF GENERATIVE AI

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Abstract

The rapid diffusion of artificial intelligence (AI), particularly generative AI (GenAI), has shifted the digital-inequality question from whether people can access digital technologies toward whether they can understand, evaluate, and use AI effectively. Yet these dimensions remain fragmented across datasets and measurement approaches. This study develops an integrated measurement framework for examining the emerging AI literacy divide across six conceptually related stages: digital access, AI access, AI literacy, demonstrated capability, effective use, and realised benefit. The framework is used as an analytical architecture rather than as an empirically demonstrated causal pathway.

The empirical analysis combines secondary-data quantitative analysis with structured cross-dataset evidence synthesis. Individual-level analysis is conducted using 1,205 observations from the FITPED AI-literacy dataset, representing 1,146 unique participants, while Eurostat, ITU, World Bank, UNESCO, OECD PIAAC, IEA ICILS, Pew Research Center, Stanford AI Index, ILO, and selected scholarly studies provide independent contextual and population-level evidence. FITPED measures perceived or self-reported AI literacy rather than demonstrated GenAI capability. The five-item AI-literacy scale has a mean of 3.991 on a five-point scale, a standard deviation of 0.676, and Cronbach's alpha of 0.688. Perceived AI literacy is positively associated with AI readiness ($\rho = .580$), relevance ($\rho = .484$), career motivation ($\rho = .450$), and behavioural intention ($\rho = .407$), and negatively associated with AI anxiety ($\rho = -.165$). Male participants and participants in IT programmes report higher AI-literacy scores than their respective comparison groups after adjustment for observed covariates. Course hours, however, do not retain an independent association with AI literacy in the exploratory regression model. A matched subset of 59 participants observed in both 2022 and 2024 shows no statistically significant systematic within-person change.

The cross-dataset evidence demonstrates that inequalities in AI use, Internet access, AI awareness, and digital capability are observable across demographic, educational, geographic, employment, and economic contexts. At the same time, these sources measure different constructs and populations and therefore cannot be interpreted as a single harmonised dataset. Emerging performance-based instruments, including GLAT and AICOS, demonstrate that objective AI and GenAI capability assessment is increasingly feasible. However, evidence remains insufficiently integrated, representative, harmonised, and longitudinal across the same individuals to connect access, demonstrated capability, effective use, and realised benefit. The study therefore identifies measurement integration—not simply access—as a central challenge for research on AI inequality. Its principal contribution is an integrated measurement framework and cross-dataset empirical synthesis that clarifies what existing evidence can establish, what it cannot establish, and where future population-level measurement should be directed.

Keywords

1. Introduction

The development and diffusion of artificial intelligence (AI) are creating a new stage in the evolution of digital inequality. Earlier digital-divide research initially concentrated on differences in access to computers, Internet connectivity, and other digital resources. Subsequent research demonstrated that access alone could not account for

differences in people's ability to use digital technologies, and later work extended the analysis toward differences in skills, uses, and the outcomes obtained from technology use (Hargittai, 2002; Robinson et al., 2015; Scheerder et al., 2017). This progression is increasingly relevant in the era of generative artificial intelligence (GenAI), where the availability of AI systems does not necessarily imply an equivalent capacity to understand, evaluate, or use them effectively.

The expansion of GenAI has therefore introduced a further layer to an already differentiated digital environment. AI systems are increasingly available to individuals through consumer applications, educational platforms, workplaces, and other digital services, but availability and use remain uneven across population groups. Recent evidence indicates differences in GenAI use associated with age, educational attainment, and gender, suggesting that access to a new generation of AI-enabled services does not automatically produce uniform patterns of participation (Suárez & García-Mariñoso, 2025). At the broader digital-access level, socioeconomic and demographic inequalities also remain consequential. The established digital-divide literature shows that technological inequalities extend beyond a simple distinction between connected and disconnected populations, encompassing differences in skills, patterns of use, and tangible outcomes (Robinson et al., 2015; Scheerder et al., 2017).

This distinction becomes particularly important for AI because interaction with an AI system requires more than technical availability. Individuals may have access to an AI-enabled service without knowing what the system can do, understanding its limitations, evaluating its outputs, recognizing inappropriate or misleading responses, or applying it effectively to a particular task. Consequently, the analytical problem is no longer adequately represented by a single access variable. A person can have access to an AI system but possess limited AI-related knowledge or confidence; another person may report substantial familiarity with AI without being able to demonstrate equivalent competence when completing an AI-supported task. These distinctions motivate the emergence of AI literacy as a distinct area of research.

AI literacy has been conceptualized as a multidimensional set of competencies rather than a single technical skill. Long and Magerko (2020), for example, conceptualized AI literacy around competencies that enable individuals to critically evaluate AI technologies, communicate and collaborate effectively with AI, and use AI as a tool across everyday and professional contexts. Ng et al. (2021) subsequently organized AI literacy around four broad aspects: knowing and understanding AI, using AI, evaluating AI, and addressing ethical issues. A systematic review by Almatrafi et al. (2024) further demonstrated the diversity of AI-literacy conceptualizations and identified recurring constructs including recognition, knowledge and understanding, use and application, evaluation, creation, and ethical navigation. Thus, AI literacy encompasses dimensions that extend beyond mere exposure to or use of AI technologies.

This multidimensionality creates an important measurement distinction. Self-reported or perceived AI literacy captures how capable individuals believe themselves to be, whereas objective or performance-based assessments seek evidence of what individuals can actually demonstrate. Chiu et al. (2024) explicitly addressed this distinction by developing and validating a 25-item objective AI-literacy test for 2,390 students in grades 7–9. Their study noted that much of the existing literature had relied on self-reported questionnaires and argued for objective assessment of AI-related knowledge and understanding. Lintner's (2024) systematic review likewise identified 22 studies validating 16 AI-literacy scales across populations including the general population, higher-education students, secondary students, and teachers, illustrating both the expansion of measurement work and the diversity of instruments being used.

The distinction between perceived literacy and demonstrated capability has become even more consequential with the emergence of GenAI. Recent assessment research demonstrates that objective GenAI-capability measurement is feasible. Jin et al. (2025), for example, developed the Generative AI Literacy Assessment Test (GLAT), a 20-item performance-based instrument validated with 355 higher-education students. GLAT scores were associated with performance on GenAI-supported tasks and provided evidence beyond self-reported perceptions of proficiency. Similarly, Markus et al. (2025) developed the AI Competency Objective Scale (AICOS), an objective measure

aligned with contemporary AI-competency models and incorporating GenAI literacy. These developments are important because they demonstrate that the distinction between perceived competence and demonstrated capability is not merely conceptual; it can increasingly be examined through different forms of measurement.

Nevertheless, AI literacy, AI capability, AI use, effective use, and realised benefit should not be treated as interchangeable constructs. Reporting that an individual uses an AI system establishes use, but does not by itself establish that the system was used competently, critically, efficiently, or appropriately. Likewise, a measure of perceived AI literacy does not establish demonstrated capability, while demonstrated capability does not necessarily establish that an individual consistently uses AI effectively in consequential contexts. Finally, effective use does not automatically establish a realised benefit. The broader digital-divide literature provides an important precedent for this distinction by differentiating access, skills and uses from the outcomes generated through technology use (Scheerder et al., 2017). The same analytical caution is necessary when examining inequalities associated with AI.

The resulting conceptual problem can be represented as a progression from the established digital divide toward increasingly specific forms of AI-related inequality:

Digital Divide → AI Access Divide → AI Literacy Divide → Capability Divide → Effective-Use Divide → Benefit Divide

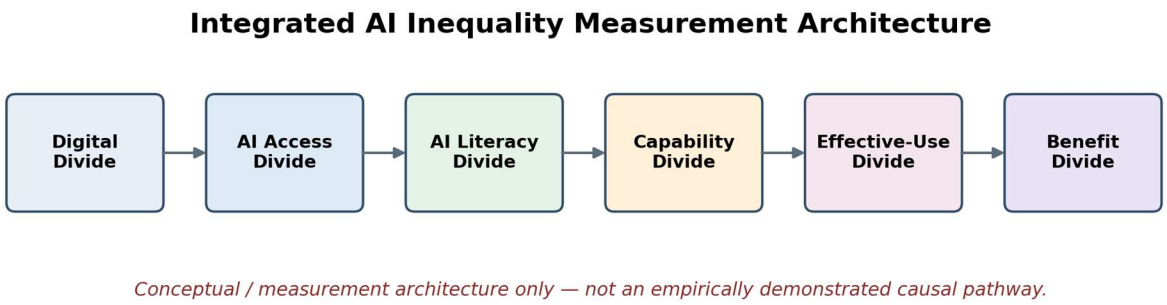


Figure 1. Integrated AI inequality measurement architecture. Conceptual/measurement architecture only; not an empirically demonstrated causal pathway.

This sequence is a conceptual measurement architecture rather than an empirically demonstrated causal chain. It provides a way to organize distinct but potentially related stages of inequality. Digital access represents an upstream structural condition. AI access concerns the availability and practical opportunity to interact with AI systems. AI literacy concerns knowledge, understanding, evaluation, use, and related competencies. Capability concerns what individuals can demonstrate rather than merely what they report. Effective use concerns the quality and appropriateness of AI-supported activity, while benefit concerns the outcomes that individuals ultimately derive from such use. The conceptual relationship among these stages is analytically important, but the existing evidence does not permit them to be treated as a single empirically observed pathway within the present study.

The measurement challenge is intensified by the fact that these dimensions are commonly observed through different instruments, populations, units of analysis, and time periods. Large-scale official statistics can provide evidence about Internet access or AI use across populations, while AI-literacy studies frequently rely on student or adult samples using self-report scales or specialized assessment instruments. International assessments such as PIAAC and ICILS provide valuable evidence concerning broader digital and computer-information capabilities, but they do not constitute measures of GenAI literacy. AI-awareness surveys provide another dimension of the emerging landscape, while occupational studies can describe potential exposure to GenAI without establishing individual capability or realised benefit. These sources are consequently informative but not directly interchangeable.

This fragmentation matters because inequalities can appear at different stages and may not have the same distribution. A population with relatively high Internet access may still contain substantial differences in AI

awareness or literacy. Similarly, high reported AI use does not necessarily imply high demonstrated capability, and high demonstrated capability does not establish that AI use produces comparable educational, occupational, economic, or social benefits. Treating these measures as if they represented the same underlying quantity risks obscuring the very inequalities that a multidimensional AI-literacy framework is intended to reveal.

The problem is particularly relevant for understanding inequalities across educational, socioeconomic, demographic, and geographic contexts. Existing evidence on digital inequalities has repeatedly demonstrated the importance of sociodemographic and socioeconomic characteristics in shaping skills, uses, and outcomes (Robinson et al., 2015; Scheerder et al., 2017). More recent research on GenAI use similarly identifies differences associated with age, education, and gender (Suárez & García-Mariñoso, 2025). At the same time, country-level and regional evidence cannot automatically be interpreted as individual-level evidence. Differences in Internet penetration, AI adoption, or awareness across countries may reflect institutional, infrastructural, economic, demographic, and cultural conditions that are not captured by an individual-level AI-literacy measure.

These considerations create a specific research problem: the emerging AI literacy divide cannot be adequately understood by asking only who has access to AI or who reports using it. It requires an examination of what is being measured at each stage, how those measures relate conceptually, and where empirical evidence is sufficiently connected to permit meaningful comparison. Importantly, this does not imply that objective GenAI-capability measurement is absent. Rather, performance-based AI and GenAI assessments are emerging, but they remain insufficiently integrated with representative measures of access, effective use, and realised benefit across the same individuals and comparable populations. The resulting limitation is therefore one of measurement integration and empirical coverage rather than the absence of capability-assessment methods.

The present study addresses this problem through a secondary-data comparative quantitative design combined with structured cross-dataset evidence synthesis. At the individual level, the study analyses the FITPED AI-literacy dataset to examine patterns and associations in perceived AI literacy within its observed student population. The study then places these findings alongside independent evidence from official and established sources concerning GenAI use, Internet access, digital capability, AI awareness, and related contextual conditions. These datasets are not merged at the respondent level, and their estimates are not treated as directly comparable measures of a single latent construct. Instead, each source is interpreted according to its population, measurement definition, unit of analysis, and temporal context.

The study has two primary contributions. First, it develops an integrated measurement framework that places access, perceived literacy, demonstrated capability, effective use, and benefit within a common analytical architecture while preserving the distinctions among them. Second, it provides a cross-dataset empirical synthesis that identifies where independent evidence converges and where measurement boundaries prevent stronger conclusions. A secondary contribution is the identification of a major measurement and data gap: although relevant measures increasingly exist for individual stages of the framework, sufficiently representative, harmonised, and longitudinal evidence linking multiple stages across the same individuals remains limited.

The study therefore does not seek to produce a new population-wide estimate of AI literacy, establish a causal model, or demonstrate an empirically complete Access → Capability → Effective Use → Benefit pathway. Instead, it asks what can currently be learned from available evidence when different forms of AI inequality are considered together without collapsing their methodological differences.

2. Conceptual Background and Literature Review

2.1 From the Digital Divide to Digital Inequality

The intellectual starting point for understanding AI-related inequality lies in the broader literature on the digital divide. Early approaches commonly conceptualised the divide in terms of unequal access to computers, Internet connections, and other information technologies. Subsequent research demonstrated that access was only one

component of digital inequality. Hargittai (2002), in her analysis of the second-level digital divide, showed that Internet users themselves differ substantially in online skills. This distinction was important because it demonstrated that technological inclusion does not imply equivalent ability to use technology.

The literature subsequently expanded the concept from a binary access problem toward a multidimensional account of digital inequality. Robinson et al. (2015) argued that digital inequalities encompass differences in access, usage, skills, and users' perceptions of their abilities. Scheerder et al. (2017), through a systematic review of research on the second- and third-level digital divide, further distinguished among determinants of Internet skills, patterns of use, and outcomes. Their review highlighted that digital-divide research had concentrated heavily on access and use, while the consequences or outcomes of technology use remained comparatively less developed.

A particularly relevant conceptual formulation was provided by Wei et al. (2011), who distinguished three levels of the digital divide: an access divide, a capability divide, and an outcome divide. In their model, access concerns inequality in access to information technology; capability concerns the ability to exploit that technology; and outcomes concern the consequences obtained from using it. Their empirical study of more than 4,000 students in Singapore provided a specific demonstration of how access, capability, and outcomes could be examined as distinct levels of inequality. The framework is particularly relevant to the present study because it establishes an intellectual precedent for treating access, capability, and outcomes as analytically distinct rather than as interchangeable indicators of digital inclusion.

The significance of this literature for AI research is methodological as much as theoretical. The progression from first-level access to second-level skills and third-level outcomes suggests that technological inequality can persist even when basic access expands. It therefore provides a useful foundation for examining whether the diffusion of AI and GenAI is producing a similar progression of inequalities. However, the present study does not assume that the digital-divide literature can simply be transferred unchanged to AI. AI introduces domain-specific competencies, evaluation requirements, ethical considerations, and interaction patterns that require separate measurement.

2.2 From Digital Access to AI Access

The expansion of AI systems has introduced another layer of technological access. Internet access remains an important upstream condition because most contemporary AI services depend on digital connectivity, devices, platforms, or institutional environments. Nevertheless, Internet access cannot be treated as an equivalent measure of AI access, and AI access cannot be treated as equivalent to AI capability.

This distinction is important because access can describe the availability of an opportunity without describing what an individual does with that opportunity. The digital-divide literature already demonstrates that access, skills, use, and outcomes can vary independently. In the AI context, the same logic applies at a more specific technological level. An individual may have Internet access and access to an AI service while lacking knowledge about AI systems, confidence in using them, or the ability to evaluate their outputs.

Recent evidence on GenAI use provides an indication that AI participation is not distributed uniformly. Suárez and García-Mariñoso (2025), analysing determinants of GenAI use, identified age, educational, and gender gaps in the use of generative AI. Such findings establish inequalities in observed use, but they do not establish inequalities in demonstrated capability or in the benefits obtained from AI use. Use is therefore best understood as a distinct empirical stage rather than as a proxy for AI literacy or effective use.

The distinction is particularly important when comparing populations with different technological, educational, or economic contexts. Evidence from high-income economies cannot automatically be interpreted as representative of developing economies, and national Internet penetration cannot by itself establish the distribution of AI capability within a population. The present study consequently treats digital access as an upstream structural condition and AI use as a separate observable dimension rather than combining them into a single measure of AI literacy.

2.3 Emergence and Conceptualisation of AI Literacy

AI literacy emerged partly in response to the recognition that individuals increasingly encounter AI systems without necessarily possessing the knowledge required to understand or critically evaluate them. Long and Magerko (2020) provided an influential conceptualisation of AI literacy as a set of competencies enabling individuals to critically evaluate AI technologies, communicate and collaborate effectively with AI, and use AI as a tool across everyday and professional contexts. Their framework positioned AI literacy as broader than technical programming knowledge.

Ng et al. (2021) subsequently examined AI literacy through an exploratory review and proposed four broad aspects: knowing and understanding AI, using AI, evaluating AI, and addressing ethical issues. This formulation reinforced the idea that AI literacy incorporates cognitive, practical, evaluative, and ethical dimensions rather than a single technical competency.

The subsequent literature has expanded this multidimensional perspective. Almatrafi et al. (2024), reviewing 47 articles published between 2019 and 2023, identified six recurring constructs: recognising AI, knowing and understanding AI, using and applying AI, evaluating AI, creating with AI, and navigating AI ethically. Their review also showed that AI-literacy research has been applied to populations ranging from school students to adults in the workforce, although educational populations represented a substantial share of the literature.

The literature therefore provides relatively strong support for treating AI literacy as multidimensional. At the same time, the multiplicity of definitions creates a measurement problem. Different studies may use the term AI literacy while measuring different combinations of knowledge, attitudes, confidence, practical skills, ethical awareness, or self-perceived competence. Consequently, two studies labelled as investigations of AI literacy may not measure the same construct to the same extent.

Zhang et al. (2025), in an umbrella review of 17 systematic reviews, similarly identified continuing needs for standardized, content-validated, and reliable AI-literacy assessments across educational levels and cultures. This observation is important for the present study because it suggests that the problem is no longer simply whether AI literacy has been conceptualised, but how consistently and comparably it can be measured.

2.4 Multidimensional Measurement of AI Literacy

The development of AI-literacy instruments has progressed rapidly, but measurement remains heterogeneous. Laupichler et al. (2023) developed the Scale for the Assessment of Non-Experts' AI Literacy (SNAIL), distinguishing dimensions related to technical understanding, critical appraisal, and practical application. Their Delphi work involved expert assessment of candidate items before preliminary validation, illustrating the movement toward construct-based measurement rather than purely ad hoc questionnaires.

Carolus et al. (2023) developed MAILS, the Meta AI Literacy Scale, drawing on competency models and psychological change- and meta-competencies. The scale incorporated dimensions including understanding, use and application, detection, ethics, and creation. Ng et al. (2024) similarly developed and validated an AI literacy questionnaire organised around affective, behavioural, cognitive, and ethical dimensions. Pinski and Benlian (2023) approached AI literacy as a multidimensional human competency incorporating socio-technical dimensions.

These instruments demonstrate that AI literacy can be operationalised in substantially different ways. Some measures emphasise technical knowledge; others include critical evaluation, practical application, attitudes, ethics, or psychological competencies. This diversity is not necessarily evidence of conceptual failure. AI literacy may genuinely be multidimensional. The methodological challenge is instead to determine which dimensions are being measured and whether results obtained using different instruments can be compared.

Lintner's (2024) systematic review provides particularly important evidence on this issue. The review identified 22 studies validating 16 AI-literacy scales across populations including general populations, higher-education students, secondary-school students, and teachers. The reviewed instruments generally demonstrated promising structural

validity and internal consistency, but relatively few had been examined comprehensively for content validity, construct validity, responsiveness, or other measurement properties. The review also reported a substantial predominance of self-report instruments over performance-based measures.

This literature establishes that AI-literacy measurement is an active and developing field. It does not, however, provide a single universally accepted instrument that can be applied across populations, countries, educational levels, and stages of the AI-use process. The measurement diversity is consequently relevant to the present study's cross-dataset approach. Rather than assuming that all AI-literacy indicators are interchangeable, the study preserves the construct boundaries of each source.

2.5 Perceived AI Literacy versus Demonstrated Capability

One of the most important distinctions emerging from the measurement literature is between perceived competence and demonstrated competence. Self-report instruments ask respondents to evaluate their own knowledge, confidence, familiarity, or capability. Such measures can be valuable for studying perceptions and attitudes, but they should not automatically be interpreted as direct evidence of performance.

Chiu et al. (2024) addressed this issue through the development and validation of an objective AI-literacy test for students in grades 7–9. Their work illustrates the movement from self-reported measures toward objective assessment. The distinction is especially relevant when an instrument asks respondents whether they know what AI can do or how confident they feel using it. Such responses provide evidence about perceived literacy, but not necessarily about whether respondents can independently demonstrate the corresponding competencies.

The distinction also appears in the emerging GenAI literature. GLAT, developed by Jin et al. (2025), is a 20-item performance-based assessment of GenAI literacy validated with 355 higher-education students. The study reported that GLAT scores predicted performance on GenAI-supported tasks more strongly than self-reported measures such as perceived ChatGPT proficiency. This provides direct evidence that perceived competence and performance-based competence can represent different empirical quantities.

The implication for AI-inequality research is substantial. A population survey may report relatively high perceived AI literacy while a performance assessment could reveal more heterogeneous capability. Conversely, low self-confidence does not necessarily demonstrate low performance. The constructs should therefore remain analytically distinct.

This distinction directly informs the interpretation of the FITPED evidence used in the present study. FITPED provides individual-level evidence concerning perceived AI literacy. It can therefore establish patterns in self-reported AI literacy and its associations with other reported constructs. It cannot establish demonstrated GenAI capability. Treating FITPED scores as objective competence would exceed what the instrument measures.

2.6 Emerging Objective GenAI-Capability Measurement

The measurement literature has moved beyond a predominantly self-report orientation. Lintner's (2024) review already documented performance-based AI-literacy measures alongside a larger number of self-report scales. More recent work has extended this development specifically into GenAI.

GLAT represents one important example. Its 20-item assessment was developed specifically for generative AI literacy and was validated using classical test theory and item-response theory with higher-education students. The study reported satisfactory reliability and structural validity and found an association between performance-based literacy scores and GenAI-supported task performance.

AICOS, developed by Markus et al. (2025), further extends objective assessment of AI literacy and incorporates GenAI literacy within an objective competency framework. The significance of such work is not that it resolves the measurement problem, but that it changes its character. Objective AI and GenAI capability can increasingly be

measured; the remaining problem concerns how such measures can be deployed consistently across populations and connected with other stages of AI participation.

This distinction is essential to the research gap proposed in the present study. It would be inaccurate to state that objective GenAI-capability measurement does not exist. The evidence instead indicates that performance-based measurement is emerging while remaining relatively fragmented in population coverage, construct scope, context, and integration with measures of access, use, and outcomes. GLAT, for example, is validated in a higher-education population rather than a nationally representative population spanning multiple socioeconomic and demographic groups.

The emerging literature therefore makes a more precise research question possible: not whether capability can be measured, but whether access, perceived literacy, demonstrated capability, effective use, and realised benefit can be observed together in sufficiently comparable populations and individuals.

2.7 AI Use versus Effective AI Use

AI use is an important but limited indicator of participation. A respondent's use of an AI system establishes that interaction occurred, but it does not necessarily reveal the quality, appropriateness, criticality, or effectiveness of that interaction.

The distinction has precedents in digital-divide research. Scheerder et al. (2017) differentiated between Internet skills, types of use, and outcomes, showing why simple measures of whether individuals use technology cannot substitute for measures of what they can do with technology or what results from that activity. Wei et al. (2011) similarly separated access, capability, and outcomes within a theoretical model of the digital divide.

The distinction becomes more consequential with GenAI because the same tool can support substantially different forms of activity. Use may range from simple information seeking to iterative problem solving, content generation, analysis, evaluation, or more complex task support. Frequency of use therefore cannot by itself establish effective use. Nor does frequent use necessarily demonstrate strong capability.

Recent GenAI-literacy research reinforces this point by connecting literacy to behaviours and performance rather than treating usage alone as the endpoint. Yan et al. (2026), for example, examined generative AI literacy in relation to privacy-protection and information-verification behaviours. Liu et al. (2025) examined GenAI literacy in relation to job performance among employees. Such studies provide evidence that literacy can be examined in relation to specific behaviours or outcomes, but they do not collectively establish a general population-level pathway from access through capability and effective use to benefit.

Accordingly, the present study treats use and effective use as conceptually distinct. Public data on GenAI use can establish patterns of adoption or participation. It cannot, without additional performance or outcome measures, establish whether individuals are using AI effectively.

2.8 AI Literacy, Education and Socioeconomic Inequality

Education occupies an important position in the literature on AI literacy and digital inequality. Many AI-literacy studies are conducted in educational settings, making students and teachers among the most frequently represented populations. Almatrafi et al. (2024) found that educational populations constituted a substantial share of the AI-literacy literature reviewed between 2019 and 2023. This concentration is useful for developing and validating instruments but also creates a limitation when attempting to generalise AI-literacy evidence to the wider adult population.

The relationship between education and digital capability also has a broader foundation in digital-inequality research. Robinson et al. (2015) conceptualised digital inequality as extending beyond access toward differences in skills and usage. Scheerder et al. (2017) similarly identified socioeconomic and demographic determinants across

different levels of the digital divide. These findings support the expectation that educational and socioeconomic context may matter for AI participation, while not establishing that education alone determines AI capability.

Recent GenAI research provides evidence of educational inequalities in use. Suárez and García-Mariñoso (2025) identified an educational gap in GenAI use. However, an educational association with use should not be interpreted as evidence that education causes AI capability or that educational attainment necessarily translates into effective use.

This distinction is particularly relevant for the present study because FITPED is itself an educational sample. Its results therefore provide evidence within a student population rather than a representative estimate of AI literacy across the general population. Cross-dataset evidence can broaden the analytical context, but it cannot transform a student sample into a population-representative estimate.

2.9 AI Adoption and Unequal Access to Benefits

The expansion of GenAI has also generated research on adoption and potential outcomes. The literature suggests that inequalities can occur not only in whether individuals encounter or use AI, but also in the capacity to convert technological interaction into useful outcomes.

Liu et al. (2025) found a relationship between GenAI literacy and job performance in an employee sample, using a multidimensional literacy construct incorporating technical competence, prompt optimisation, content evaluation, innovative application, and ethics/compliance. This provides evidence that GenAI literacy can be studied in relation to an outcome that is relevant to the effective-use and benefit dimensions of the proposed framework. It does not, however, demonstrate a general causal pathway from access to job outcomes.

Similarly, Yan et al. (2026) examined relationships between generative AI literacy and privacy-protection and information-verification behaviours. Such research indicates that literacy can have behavioural implications, but the existence of an association in a selected study population should not be extrapolated to all AI users or interpreted as proof of a universal benefit mechanism.

At the macro level, the literature also demonstrates unequal AI adoption across economic contexts. The World Bank's work on generative AI adoption identifies substantial differences in consumer-facing GenAI engagement across income groups. Such evidence is useful for understanding structural inequalities in adoption, but traffic and platform-use measures are not equivalent to representative measures of individual capability, effective use, or realised benefit.

The implication is that adoption should be treated as an intermediate empirical phenomenon rather than as a direct measure of benefit. An individual may adopt AI without obtaining substantial benefits, while another may obtain substantial benefits from relatively limited but highly effective use. Measuring these distinctions requires individual-level outcome data that are currently much less common than access or use measures.

2.10 India and the Global South: Evidence Boundaries

The geographic distribution of AI-literacy research creates an additional interpretive problem. Much of the established measurement literature has been developed in educational or adult populations in particular national settings, and evidence from high-income countries cannot automatically be treated as representative of developing countries or the Global South.

Selected evidence from India illustrates the importance of maintaining this distinction. Studies of ChatGPT adoption among higher-education students in India have examined technology-acceptance factors and patterns of use, including studies involving students in Punjab and northern Indian states. These studies provide useful evidence concerning selected Indian student populations, but they are not nationally representative estimates of AI literacy or demonstrated GenAI capability. (Singh et al., 2025; Jasrai, 2025).

Similarly, research from Sri Lanka has examined barriers and opportunities for GenAI adoption in higher education in the Global South. Such evidence is valuable for understanding contextual conditions affecting adoption, but a single-country higher-education study should not be generalized to the entire Global South. (Henadirage & Gunarathne, 2025).

The broader structural evidence nevertheless indicates that access conditions remain unequal within developing-country contexts. For India, Internet-access statistics provide evidence of differences between urban and rural populations and between men and women. These indicators establish an access inequality, but they do not measure AI literacy or GenAI capability. The analytical distinction is therefore essential: structural digital inequality can constrain opportunities for AI participation without directly measuring the competencies required for effective AI use.

The present study accordingly uses India and Global South evidence as contextual and structural evidence rather than treating it as interchangeable with FITPED or with population estimates from high-income countries. This approach prevents the common methodological error of using a geographically narrow AI-literacy study as though it represented a national or global population.

2.11 Measurement Fragmentation Across the Literature

The literature reviewed above reveals a field that has developed rapidly but along multiple measurement trajectories.

The first trajectory concerns **access**. Digital-divide research and official statistics can measure connectivity, technology availability, and participation in digital environments. Such measures are important because access represents an upstream condition, but access does not establish capability.

The second concerns **perceived AI literacy**. A growing body of AI-literacy scales measures self-reported knowledge, confidence, attitudes, use, evaluation, ethics, or related competencies. These instruments have contributed substantially to conceptual development, but their constructs are heterogeneous and their populations vary.

The third concerns **demonstrated capability**. Objective and performance-based AI-literacy assessment is increasingly available, including instruments developed for school and higher-education populations. GLAT and AICOS illustrate the recent expansion of this trajectory into GenAI-specific or GenAI-inclusive assessment.

The fourth concerns **use**. Surveys and adoption studies can establish whether and how frequently people interact with AI systems. Such evidence is valuable for measuring participation, but it does not necessarily establish effective use.

The fifth concerns **behavioural and substantive outcomes**. Emerging studies connect GenAI literacy with behaviours such as information verification or privacy protection and with outcomes such as job performance. These studies provide important evidence that AI literacy can have consequences beyond self-perception, but their populations, measures, and outcomes remain study-specific.

The sixth concerns **realised benefit**. Benefit is conceptually distinct from use and capability, yet it is considerably more difficult to measure. Educational, occupational, economic, or social benefits may depend on the context in which AI is used and on factors beyond individual literacy. Consequently, evidence on realised individual benefit remains less integrated with the other dimensions of AI inequality.

This fragmentation means that the literature contains many pieces of the broader problem without routinely observing all pieces within the same individuals. The consequence is not that the individual literatures are invalid. Rather, each provides evidence about a particular stage or construct, while the relationships between stages often remain only partially observable.

2.12 Integrated Literature Synthesis

Taken together, the literature suggests that the emerging AI literacy divide should not be understood as a single difference between people who have AI and people who do not. The evidence instead points toward multiple dimensions of participation and competence.

The first dimension is **structural access**. Without connectivity, devices, platforms, or institutional access, individuals may have limited opportunities to interact with AI. Yet access alone does not determine what individuals can do with the technology.

The second dimension is **AI awareness and exposure**. Individuals must encounter or recognise AI systems before meaningful engagement can occur. Awareness, however, is not equivalent to literacy or capability.

The third dimension is **perceived AI literacy**. Individuals may report different levels of knowledge, confidence, understanding, or familiarity. Such measures are valuable for examining perceptions and self-assessments, but they should not be interpreted as direct measures of performance.

The fourth dimension is **demonstrated capability**. Emerging objective instruments show that AI and GenAI competencies can increasingly be assessed through performance-based methods. This represents an important development in the field, but existing instruments remain bounded by their populations, contexts, and validation designs.

The fifth dimension is **effective use**. Capability is not necessarily equivalent to consistent, appropriate, or productive use. Effective use requires attention to task context, output evaluation, critical judgement, and the purposes for which AI is employed.

The sixth dimension is **realised benefit**. The consequences of AI use may include educational, occupational, informational, economic, or other outcomes. These outcomes are shaped by both individual capability and contextual conditions, making them difficult to attribute to AI literacy alone.

The conceptual movement from digital divide to AI access, literacy, capability, effective use, and benefit therefore provides a useful analytical structure, but the literature does not justify treating these stages as a single empirically demonstrated pathway. The strongest conclusion supported by the existing evidence is that the stages are related conceptual dimensions that require better empirical integration.

This synthesis also clarifies why the term **AI literacy divide** should be interpreted carefully. It does not denote a single scalar gap that can be measured through one universal instrument. It refers to a broader pattern of unequal opportunities, knowledge, perceived competence, demonstrated capability, use, and potential outcomes associated with AI systems. The precise composition of that divide depends on the measurement instrument and population under examination.

3. Research Gap, Questions and Analytical Objectives

3.1 What the Literature Already Establishes

The literature supports several conclusions with reasonable confidence.

First, digital inequality is multidimensional. Research has moved beyond a simple access/no-access distinction toward skills, usage, capability, and outcomes (Hargittai, 2002; Robinson et al., 2015; Scheerder et al., 2017; Wei et al., 2011).

Second, AI literacy is an established and increasingly multidimensional research construct. Foundational conceptualisations identify competencies involving understanding, use, evaluation, communication, creation, and ethical engagement with AI (Long & Magerko, 2020; Ng et al., 2021; Almatrafi et al., 2024).

Third, AI-literacy measurement has expanded substantially. Multiple self-report and performance-based instruments have been developed for different populations and purposes (Laupichler et al., 2023; Carolus et al., 2023; Chiu et al., 2024; Lintner, 2024; Ng et al., 2024).

Fourth, objective measurement of AI and GenAI capability is emerging. GLAT and AICOS demonstrate that performance-based assessment is increasingly feasible, and GLAT provides evidence that performance-based GenAI literacy can relate to task performance more directly than self-reported proficiency in its study population.

Fifth, inequality in AI use is observable. Research has identified differences in GenAI use associated with age, education, and gender, while broader digital-inequality research demonstrates that access and capability are unequally distributed (Suárez & García-Mariñoso, 2025; Robinson et al., 2015; Scheerder et al., 2017).

Sixth, selected studies have begun connecting AI or GenAI literacy with behaviours and outcomes, including information verification, privacy-related behaviour, and job performance (Yan et al., 2026; Liu et al., 2025). These studies demonstrate that outcome-oriented AI-literacy research is possible, but they do not establish a universal causal sequence.

3.2 What Remains Fragmented

The narrower unresolved problem concerns integration.

Existing research frequently observes one or more stages of AI-related inequality but does not routinely observe access, perceived literacy, demonstrated capability, effective use, and realised benefit together within the same individuals and comparable populations. Measurement instruments differ in constructs, item formats, scoring, target populations, geographic contexts, and levels of analysis. Performance-based measures are emerging, but many have been validated in specific educational populations rather than representative national samples.

This creates several discontinuities.

Construct discontinuity occurs when measures labelled as AI literacy assess different combinations of knowledge, attitudes, confidence, practical application, evaluation, or ethics.

Measurement-mode discontinuity occurs when one study uses self-report while another uses objective performance assessment.

Population discontinuity occurs when evidence comes from students, teachers, employees, general adults, or other selected populations that cannot automatically be compared.

Geographic discontinuity occurs when evidence from high-income countries is compared with evidence from developing countries or India without accounting for different structural contexts.

Stage discontinuity occurs when access, use, capability, or outcomes are measured separately rather than together.

Temporal discontinuity occurs when different stages are measured in different years, making apparent differences difficult to interpret as changes within the same population.

These discontinuities limit the extent to which the existing literature can support a complete empirical account of the proposed conceptual movement:

Digital Divide → AI Access Divide → AI Literacy Divide → Capability Divide → Effective-Use Divide → Benefit Divide

The literature therefore supports the conceptual importance of connecting these stages, but it does not establish that they form a universally observed causal chain.

3.3 The Precise Research Gap

The research gap addressed by this study is consequently narrower than the claim that AI literacy or AI capability has been neglected.

AI literacy has been extensively conceptualised and increasingly measured. AI capability has also begun to be measured objectively. GenAI-specific performance assessment is emerging. Inequalities in AI use have been documented, and selected studies have examined relationships between AI literacy and behavioural or occupational outcomes.

The remaining gap is that **these forms of evidence remain insufficiently integrated across access, perceived literacy, demonstrated capability, effective use, and realised benefit within the same individuals and comparable populations**.

This is a measurement and evidence-integration gap rather than an absence-of-literature gap.

The distinction is important. The problem is not that researchers lack all of the necessary instruments. Rather, the instruments and datasets have developed along partially separate trajectories. Consequently, it remains difficult to determine how inequalities observed at one stage correspond to inequalities at another stage, whether particular populations move from access to demonstrated capability, whether capability translates into effective use, and whether effective use produces comparable benefits.

The gap is especially consequential for population-level research. A performance-based instrument validated in a higher-education sample can establish that capability is measurable in that context, but it cannot by itself provide a nationally representative estimate. A national survey of AI use can establish participation patterns, but it cannot by itself establish capability. An Internet-access indicator can establish structural connectivity, but it cannot establish AI literacy. An occupational outcome study can identify an association between literacy and job performance, but it cannot reconstruct the upstream pathway from access.

The research challenge is therefore to place these forms of evidence into a common analytical architecture while preserving their methodological differences.

3.4 Positioning of the Present Study

The present study responds to this gap through two primary contributions.

The first is an **integrated measurement framework**. The framework organises evidence across access, perceived AI literacy, demonstrated capability, effective use, and benefit while maintaining explicit distinctions among these constructs. It is intended as a measurement architecture that can guide future empirical integration, not as a newly validated causal model.

The second is a **cross-dataset empirical synthesis**. The study combines direct individual-level analysis of the FITPED AI-literacy dataset with independent evidence from official international statistics, established assessments, AI-awareness research, adoption evidence, and relevant scholarly studies. The datasets are not merged at the respondent level. Instead, each source is interpreted according to its own construct, population, year, and measurement design.

A secondary contribution is the **identification of a major measurement/data gap**. The synthesis demonstrates where evidence already exists and where the connection between stages remains incomplete. In particular, the study identifies the need for sufficiently representative, harmonised, and longitudinal data capable of observing multiple stages of AI inequality within the same individuals or suitably comparable populations.

The study therefore does not claim to provide a new population-wide estimate of AI literacy, establish causality, or demonstrate an empirically complete Access → Capability → Effective Use → Benefit pathway. Its contribution lies in clarifying the empirical architecture of the AI literacy divide and specifying the measurement conditions required for stronger future tests.

3.5 Implication for the Present Research Design

This gap directly motivates the research design adopted in the present study. Because no single available dataset in the assembled evidence base simultaneously measures all relevant stages at the required level of comparability, the study does not attempt to construct a synthetic individual-level pathway by combining incompatible datasets.

Instead, the empirical strategy deliberately separates:

1. **Direct individual-level evidence**, obtained from FITPED, which measures perceived AI literacy and related self-reported constructs;
2. **Population-level and structural evidence**, including official measures of GenAI use, Internet access, digital capability, and AI awareness;
3. **Capability-assessment evidence**, including broader digital or computer-information assessments and emerging objective AI/GenAI-literacy instruments;
4. **Outcome-oriented evidence**, where selected studies examine behavioural or occupational consequences of AI literacy.

The resulting synthesis does not resolve the full pathway. Rather, it identifies which links can currently be observed and which require future integrated data.

This approach is consistent with the broader development of the digital-divide literature. As earlier research moved from access toward capability and outcomes, AI research now requires an analogous but domain-specific expansion from AI availability and use toward competence, effective use, and benefit. The present study therefore treats the AI literacy divide as both an inequality problem and a measurement problem.

3.6 Research Questions

Accordingly, the study addresses four research questions:

RQ1. How does perceived AI literacy vary within FITPED and how is it associated with related AI attitudes, readiness, and behavioural intention?

RQ2. What inequality patterns in GenAI use, digital access, and AI awareness are observable across age, education, geography, gender, employment, and economic context in independent population datasets?

RQ3. How do public datasets operationalise access, perceived AI literacy, digital capability, GenAI use, and AI-related outcomes, and where are the principal measurement discontinuities?

RQ4. To what extent can existing evidence support an integrated framework linking access, capability, effective use, and benefit?

3.7 Analytical Positioning

Together, these questions position the AI literacy divide not as a single measurable gap, but as a multidimensional empirical and measurement problem. The central task is consequently to determine which forms of inequality are already observable, which remain only partially measured, and where the evidence base requires stronger integration before relationships across the stages of the proposed framework can be empirically tested.

4. Methodology

4.1 Study Design

The study uses a **secondary-data comparative quantitative analysis combined with structured cross-dataset evidence synthesis**.

The design has two complementary components.

The first is an individual-level quantitative analysis of the FITPED AI Literacy Questionnaire dataset. FITPED provides the principal empirical dataset in the study and permits examination of perceived AI literacy, its constituent items, related self-reported constructs, demographic and programme differences, longitudinal observations for a small matched subset, correlations, and an exploratory multivariable regression.

The second component is a structured synthesis of independent official, international, and scholarly evidence. Eurostat, ITU, World Bank WDI, UNESCO, PIAAC, ICILS, Pew Research Center, Stanford AI Index, ILO, and verified World Bank/Semrush research provide evidence concerning different aspects of AI use, digital access, digital capability, AI awareness, adoption, and occupational exposure.

These sources are deliberately treated as **independent evidence streams rather than as components of one pooled statistical dataset**.

Respondent-level merging was not performed because the sources differ substantially in population, sampling framework, geographic coverage, year, instrument, unit of analysis, and construct definition. For example, FITPED provides individual-level self-reported AI-literacy observations, whereas Eurostat provides population-level GenAI-use indicators, ITU provides structural digital-access indicators, and ICILS and PIAAC provide assessment-based evidence concerning broader digital or computer-information capabilities.

The design therefore prioritises **measurement transparency over artificial statistical integration**. Combining incompatible datasets into a synthetic individual-level model would imply comparability that the underlying evidence does not provide.

4.2 Data Sources

4.2.1 FITPED AI Literacy Questionnaire Dataset

The FITPED AI Literacy Questionnaire dataset is the primary individual-level empirical source (Skalka et al., 2025a; Skalka et al., 2025b).

The dataset contains **1,205 survey records representing 1,146 unique participants**. There are **59 repeated participant IDs**, corresponding to participants with observations in both 2022 and 2024. No exact duplicate rows were identified.

The dataset contains the principal AI-literacy measure used in the study. The five AI-literacy items are:

- L1: AI can be used for image recognition/search.
- L2: AI can be used for speech recognition/search.
- L3: (will) use AI-assisted online translation.
- L4: (will) communicate with an AI voice assistant.
- L5: AI can predict some things such as music/books.

The questionnaire documentation establishes these as the AI-literacy items used in the analysis; the instrument was adapted from the validated questionnaire of Dai et al. (2020), as documented in the FITPED data article (Skalka et al., 2025a).

The questionnaire additionally contains constructs concerning relevance, career motivation, confidence, social goods, intrinsic motivation, anxiety, readiness, and behavioural intention.

FITPED is therefore the only dataset in the study used for the principal individual-level statistical analysis.

4.2.2 Eurostat

Eurostat provides independent population-level evidence concerning GenAI use among individuals aged 16–74 in the European Union.

The relevant Eurostat evidence measures GenAI use across dimensions including age, education, geographic location, gender, employment status, and country. It therefore serves as contextual evidence for the **AI-use and inequality** components of the conceptual framework.

Eurostat is not treated as a measure of demonstrated AI capability or realised individual benefit.

4.2.3 ITU DataHub

The International Telecommunication Union provides structural evidence concerning digital access.

For the present study, ITU evidence is used primarily to contextualise Internet access and internal access inequalities, including urban-rural and gender differences in India.

These indicators operationalise an **upstream digital-access condition**, not AI literacy or demonstrated GenAI capability.

4.2.4 World Bank World Development Indicators

The World Bank World Development Indicators provide corroborating evidence for the Internet-use indicator IT.NET.USER.ZS.

The indicator measures individuals using the Internet as a percentage of the population and is based on the ITU statistical system. Consequently, WDI is treated as a corroborating source within the same indicator family rather than as an entirely independent survey estimate.

Its role is therefore structural and contextual: it supports interpretation of Internet access and digital participation rather than directly measuring AI literacy.

4.2.5 UNESCO Institute for Statistics

UNESCO UIS provides the broader digital-literacy framework and measurement context.

Its digital-literacy framework defines digital literacy in terms of confident and critical use of digital technologies for information, communication, and basic problem solving. The relevant SDG indicator concerns attainment of minimum digital-literacy proficiency.

This evidence is used to distinguish **broader digital literacy from GenAI literacy**. UNESCO evidence therefore contributes to the capability context without being interpreted as a direct measure of GenAI capability.

4.2.6 OECD PIAAC

The OECD Programme for the International Assessment of Adult Competencies provides assessment-based evidence concerning adult skills, including digital/problem-solving capabilities.

PIAAC Cycle 2 Round 1 involved 31 countries/economies, with data collection conducted between September 2022 and August 2023. India was not among the Round 1 participating countries.

PIAAC is used as a **capability precedent and contextual assessment source**. It is not treated as a GenAI-literacy dataset.

4.2.7 IEA ICILS

The International Computer and Information Literacy Study provides international assessment evidence concerning computer and information literacy.

ICILS 2023 covered 35 education systems and more than 130,000 students. Its assessment concerns Computer and Information Literacy rather than GenAI literacy.

Accordingly, ICILS is used to demonstrate the availability of large-scale performance-based evidence for broader digital capabilities while maintaining the distinction between such capabilities and GenAI-specific competence.

4.2.8 Pew Research Center

Pew Research Center evidence provides contextual information about public awareness of AI.

The evidence is used to distinguish **AI awareness** from AI literacy and capability. Awareness that AI exists or has been discussed does not establish that an individual understands AI systems or can use them effectively.

Pew evidence therefore occupies an intermediate contextual position in the framework rather than serving as a direct capability measure.

4.2.9 Stanford AI Index

The Stanford AI Index provides broader contextual evidence concerning AI adoption and diffusion.

Its role is to establish the scale and pace of AI adoption within the wider technological environment. It is not used as an individual-level AI-literacy measure and is not treated as evidence of demonstrated capability.

4.2.10 International Labour Organization

ILO evidence provides an occupational perspective on potential exposure to GenAI.

The ILO's global analysis identifies occupations and employment shares potentially exposed to GenAI. Exposure is treated as a contextual labour-market condition rather than evidence of individual AI capability, AI use, productivity gain, or realised benefit.

4.2.11 World Bank/Semrush GenAI-Adoption Evidence

Verified World Bank research using Semrush data provides cross-income-group evidence concerning consumer-facing GenAI adoption and traffic patterns.

This evidence is used to contextualise structural differences in GenAI adoption across income groups. Traffic and platform-use measures are not interpreted as equivalent to representative individual-level prevalence, demonstrated capability, effective use, or realised benefit.

Table 1. Dataset and Evidence Inventory

Source	Primary construct / evidence	Level	Role in this study
FITPED AI Literacy Questionnaire dataset	Perceived AI literacy; related attitudes/readiness/intentions	Individual	Primary empirical dataset
Eurostat (isoc_ai_iaiu)	Reported GenAI use and inequality dimensions	Population	GenAI-use context
ITU DataHub	Internet access/use and India disaggregation	Population / household	Digital-access context
World Bank WDI	Individuals using the Internet (IT.NET.USER.ZS)	Population	Corroborating access indicator
UNESCO UIS	Digital-literacy framework / proficiency context	Framework / population	Broader capability context

OECD PIAAC	Adult skills and problem-solving capability	Adult assessment	Performance-based capability context
IEA ICILS	Computer and Information Literacy	School assessment	Performance-based digital-capability context
Pew Research Center	AI awareness	Population survey	Awareness context
Stanford AI Index	AI adoption/diffusion context	Aggregate	Adoption context
International Labour Organization	Potential occupational GenAI exposure	Occupational / aggregate	Labour-market context
World Bank/Semrush	Consumer-facing GenAI adoption/traffic	Aggregate	Income-group adoption context
GLAT / AICOS and related studies	Objective AI/GenAI capability measurement	Selected samples	Emerging capability evidence

Source: Locked Step 5 evidence matrix and verified source set.

Note: The evidence streams are not respondent-level merged and are not treated as measurements of one common construct.

4.3 Sample and Data Description

4.3.1 FITPED Sample Structure

The FITPED dataset contains **1,205 records from 1,146 unique participants**.

The difference between records and unique participants is attributable to **59 repeated participant IDs**. These repeated observations correspond to participants observed in both 2022 and 2024 and permit a limited matched-subset analysis.

There were **no exact duplicate rows**.

There were **no raw missing cells** in the supplied dataset.

The original data were not overwritten during processing.

4.3.2 Treatment of Zero-Coded Non-Applicable Values

Although the raw dataset contained no empty cells, zero values occurred in several constructs as non-applicable/missing-coded responses.

Zeros were therefore treated as missing for scale construction for the relevant constructs. This treatment was applied to **IM, S, C, and BI**.

The AI-literacy items themselves use the observed 0–5 response scale and were not transformed on the basis of the zero-coded treatment applied to those other constructs.

4.3.3 AI-Literacy Scale

The primary AI-literacy construct consists of five items, L1–L5.

Each item is measured on a 0–5 scale. The AI-literacy score was constructed as the **mean of valid items**, with a minimum of **80% valid items** required for scale calculation.

Thus, the five-item AI-literacy scale retains the original instrument structure rather than deleting individual items to improve internal consistency.

The resulting FITPED AI-literacy scale has:

- **N = 1,205**
- **Mean = 3.991**
- **SD = 0.676**
- **Median = 4.0**
- **Cronbach's α = 0.688**

The individual item means were:

- L1 = **4.559**
- L2 = **4.529**
- L3 = **3.954**
- L4 = **3.230**
- L5 = **3.684**

L4 showed a comparatively weaker contribution to internal consistency. Removing L4 would increase alpha to approximately 0.743; however, L4 was retained because it is an original component of the validated questionnaire structure used for the present analysis. The resulting alpha is therefore interpreted as moderate internal consistency rather than as grounds for altering the original instrument.

4.3.4 Other FITPED Constructs

The available FITPED constructs and their locked descriptive/reliability statistics were:

The different sample sizes reflect the availability of valid observations after applying the locked treatment of non-applicable/missing-coded responses.

Analytical-N clarification: the effective analytical Ns in this study are based on the prespecified ≥80% valid-item rule and may therefore differ from the effective Ns reported in the original FITPED data article, which follows its own dataset-reporting convention.

The related FITPED construct statistics are retained as follows: AI Readiness (N = 1,205, Mean = 3.827, SD = 0.687, α = .825); Relevance (N = 1,205, Mean = 3.797, SD = 0.699, α = .828); Social Goods (N = 1,205, Mean = 3.775, SD = 0.682, α = .736); Career Motivation (N = 1,205, Mean = 3.666, SD = 0.860, α = .848); AI Anxiety (N = 1,205, Mean = 2.577, SD = 0.937, α = .882); Intrinsic Motivation (N = 634, Mean = 3.536, SD = 0.830, α = .832); Satisfaction (N = 684, Mean = 3.517, SD = 0.774, α = .881); Confidence (N = 816, Mean = 3.584, SD = 0.743, α = .865); Behavioural Intention (N = 989, Mean = 3.775, SD = 0.781, α = .878).

4.4 Operationalisation

The study deliberately distinguishes constructs that are frequently conflated in discussions of AI inequality.

This operationalisation prevents the study from substituting one construct for another.

In particular:

Internet access is not AI literacy.

AI awareness is not AI capability.

Perceived AI literacy is not demonstrated capability.

AI use is not necessarily effective AI use.

Effective use is not equivalent to realised benefit.

Education is not itself a measure of AI capability.

These distinctions are central to the interpretation of the cross-dataset synthesis.

4.5 Data Processing

The FITPED data were subjected to structured data-quality and construct-processing procedures before statistical analysis.

The initial audit established the number of records and unique participants, identified repeated participant IDs, checked for exact duplicate rows, examined raw missingness, and reviewed the observed response ranges.

The five AI-literacy items were retained in their original structure. Scale scores were calculated using the mean of valid items with a minimum 80% validity criterion.

For constructs containing zero-coded non-applicable responses, zeros were treated as missing according to the locked Step 5 processing decision. The original raw dataset was not overwritten.

The 59 repeated participants were retained as a matched longitudinal subset for a separate within-person comparison between their 2022 and 2024 observations.

No attempt was made to convert the independent aggregate datasets into individual-level observations.

No composite index combining Internet access, AI literacy, capability, use, and benefit was constructed.

No respondent-level linkage between FITPED and Eurostat, ITU, WDI, UNESCO, PIAAC, ICILS, Pew, Stanford, ILO, or World Bank/Semrush data was attempted.

4.6 Statistical Analysis

The FITPED analysis comprises the following statistical procedures.

4.6.1 Descriptive Statistics

Descriptive statistics were calculated for the AI-literacy scale and the available related constructs.

The analysis reports sample size, mean, standard deviation, and median where applicable.

Item-level AI-literacy distributions were also examined to identify differences among L1–L5.

4.6.2 Reliability Analysis

Internal consistency of multi-item constructs was assessed using **Cronbach's alpha**.

For AI literacy, the five original items were evaluated as a scale. The analysis identified the comparatively lower contribution of L4 and calculated the effect of removing L4 as a sensitivity observation, but the item was retained in the principal analysis.

Reliability statistics were also calculated for the other FITPED constructs.

4.6.3 Group Comparisons

AI-literacy scores were compared across relevant demographic and educational groups.

Gender results were:

- Male: **N = 793, Mean = 4.113**
- Female: **N = 402, Mean = 3.752**
- Other: **N = 10, Mean = 3.900**

The unadjusted male–female difference was approximately **0.361**, corresponding to an unadjusted standardized difference of approximately **d = 0.55**.

Programme results were:

- IT: **N = 832, Mean = 4.107**
- Education: **N = 188, Mean = 3.730**
- Other: **N = 68, Mean = 3.794**
- Management: **N = 52, Mean = 3.796**
- Language: **N = 28, Mean = 3.471**
- IT education: **N = 22, Mean = 3.891**
- STEM education: **N = 14, Mean = 3.471**
- it: **N = 1, Mean = 4.400**

The raw IT–Education difference was approximately **0.377**, with an unadjusted standardized difference of approximately **d = 0.57**.

Year-specific descriptive results were:

- 2022: **N = 535, Mean = 4.084**
- 2023: **N = 112, Mean = 4.134**
- 2024: **N = 558, Mean = 3.873**

Because participant composition differs across years, these figures are not interpreted as a causal temporal trend.

Country-level descriptive results were:

- Slovakia (SK): N = 686, Mean = 3.861
- Poland (PL): **N = 299, Mean = 4.233**
- Czech Republic (CZ): **N = 106, Mean = 3.987**
- Indonesia (ID): **N = 65, Mean = 4.046**
- Lithuania (LT): **N = 38, Mean = 4.253**
- Türkiye (TR): **N = 6, Mean = 4.567**
- France (FR): **N = 3, Mean = 3.667**
- Ukraine (UA): **N = 2, Mean = 4.800**

These country figures are descriptive only. Country and year are partly confounded: Indonesia and Lithuania are represented only in 2023, while the 2022 and 2024 observations are concentrated in Slovakia, Poland, and the Czech Republic. The results therefore do not support causal country comparisons.

4.6.4 Matched Longitudinal Subset

The repeated-ID structure permits a limited within-person analysis of the **59 participants observed in both 2022 and 2024**.

For this matched subset:

- 2022 mean = **4.044**

- 2024 mean = **3.898**
- Mean within-person change = **−0.146**
- 24 participants increased
- 26 participants decreased
- 9 participants were unchanged

The paired Wilcoxon analysis produced **p = .324**, with a rank-biserial effect of approximately **−.159**.

The analysis therefore provides no statistically significant evidence of a systematic within-person change in perceived AI literacy in this matched subset.

Given the small matched sample, this result is not interpreted as evidence about population-wide change.

4.6.5 Correlation Analysis

Spearman rank correlations were calculated between AI literacy and the available related FITPED constructs.

The correlations were:

These coefficients are interpreted as associations only.

4.6.6 Exploratory Regression

An exploratory ordinary least squares regression examined AI literacy as the dependent variable.

The model included:

- year indicators,
- age,
- gender,
- IT-programme indicator,
- log(1 + AI-course hours).

Cluster-robust standard errors were used at the participant (user_id) level.

The model explained **R^2 = .090** of the observed variance.

The coefficients were:

The adjusted association between AI-course hours and AI literacy is therefore not statistically significant in this model, despite a small positive bivariate association between course hours and AI literacy (**Spearman ρ = .104, p < .001**).

The year coefficient for 2024 should not be interpreted as evidence that AI literacy declined over time in the population. The year variable is partly entangled with changes in sample composition, and the separate matched-subset analysis did not identify statistically significant systematic within-person change.

The gender and IT-programme coefficients indicate adjusted associations within the observed FITPED sample. They do not establish causal effects.

4.6.7 Regression Sensitivity

The analysis included sensitivity consideration around the AI-literacy scale's internal consistency and model specification.

In particular, L4 was identified as the item with the weaker item-total contribution, and removal would raise Cronbach's alpha to approximately **.743**. Nevertheless, L4 was retained in the primary scale because removing an original instrument item would change the construct being analysed.

The regression also demonstrated that the small bivariate association between AI-course hours and AI literacy does not persist after adjustment for the included covariates. This reinforces the need to distinguish descriptive association from adjusted association.

4.7 Cross-Dataset Evidence Synthesis and Comparability

The cross-dataset component does not statistically pool estimates from the different sources.

Instead, each source is classified according to:

1. **Construct measured**
2. **Measurement level**
3. **Population**
4. **Geographic coverage**
5. **Year or observation period**
6. **Sampling framework**
7. **Instrument or indicator definition**
8. **What the source can establish**
9. **What the source cannot establish**

The synthesis then examines convergence and discontinuity across these independent evidence streams.

For example, Eurostat can establish population-level inequalities in reported GenAI use across demographic and socioeconomic categories. ITU and WDI can establish structural differences in Internet access. UNESCO, PIAAC, and ICILS can contextualise broader digital capability. Pew can provide evidence concerning AI awareness. GLAT and AICOS demonstrate the emerging feasibility of objective AI/GenAI-capability assessment. ILO can provide occupational exposure context, while World Bank/Semrush evidence can contextualise differences in consumer-facing GenAI adoption across economic groups.

These observations are not treated as measurements of the same construct.

The synthesis therefore asks whether the available evidence collectively covers the conceptual stages of the proposed framework and where discontinuities prevent stronger integration.

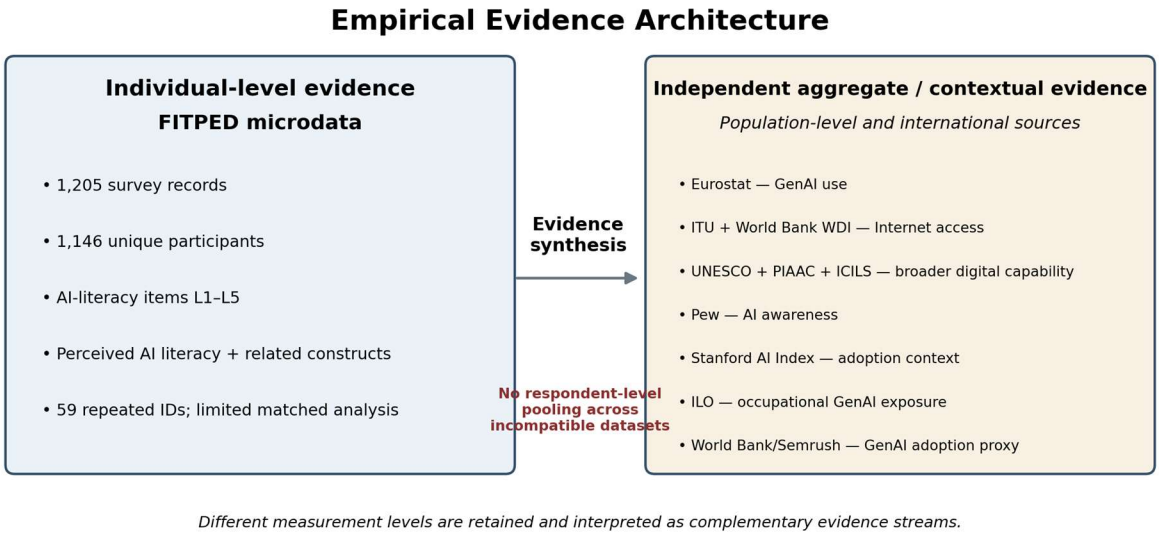


Figure 2. Empirical evidence architecture. FITPED provides individual-level microdata; the remaining evidence streams are independent aggregate or contextual sources and are not pooled at respondent level.

4.7.1 Cross-Dataset Comparability

Respondent-level merging was explicitly avoided because the datasets are not harmonised at the individual level.

The principal differences include:

Population differences. FITPED consists of a student-oriented sample, whereas Eurostat covers individuals aged 16–74 in the relevant European statistics, PIAAC concerns adults, ICILS concerns school students, and Pew surveys adults across selected countries.

Temporal differences. The sources cover different years and observation periods. Consequently, differences between datasets cannot automatically be interpreted as temporal change.

Instrument differences. FITPED uses a self-reported AI-literacy questionnaire; Eurostat measures GenAI use; ITU measures Internet access; PIAAC and ICILS use assessment frameworks for broader skills; Pew measures awareness; and objective instruments such as GLAT measure GenAI literacy through performance-oriented assessment.

Sampling differences. Some sources are population surveys, some are educational assessments, some are selected research samples, and some are administrative or traffic-derived indicators.

Measurement-level differences. FITPED is individual-level, while several official sources provide aggregate or population-level estimates. These levels cannot be combined as if they represented equivalent observations.

Geographic differences. Sources cover different countries and regions, with some providing international evidence and others focusing on specific national or educational populations.

For these reasons, cross-dataset synthesis is used to identify patterns and measurement discontinuities rather than to calculate a single pooled effect or construct an individual-level causal pathway.

4.8 Research Integrity and Ethics

The study is based on secondary data and publicly available or documented sources.

The FITPED dataset is analysed in its supplied form and is treated as the principal individual-level empirical dataset. The analysis does not claim ethical approval because no such approval is documented within the locked research architecture.

Official statistical sources and established international assessments are used with attention to their published definitions, populations, years, sampling frameworks, and measurement limitations.

Research integrity is maintained through several principles:

- preservation of the original FITPED observations;
- explicit documentation of missing/non-applicable treatment;
- retention of the original AI-literacy item structure;
- separation of individual-level and aggregate evidence;
- no respondent-level merging of incompatible datasets;
- explicit distinction between self-report and demonstrated performance;
- explicit distinction between use and effective use;
- citation and traceability of external evidence;
- avoidance of unsupported causal interpretation;
- transparent reporting of sample and measurement limitations.

The analysis therefore aims to make the evidentiary boundaries of each source visible rather than conceal them through statistical aggregation.

5. Results and Empirical Findings

5.1 FITPED Dataset and Analytical Sample

The FITPED dataset contained 1,205 survey records from 1,146 unique participants. The difference between the number of records and unique participants resulted from 59 repeated participant IDs. These repeated observations corresponded to participants observed in both 2022 and 2024 and were subsequently used for a limited matched-subset analysis. (Skalka et al., 2025a; Skalka et al., 2025b).

No exact duplicate rows were identified. The raw dataset contained **no missing cells**. However, zero-coded non-applicable responses occurred in the IM, S, C, and BI constructs and were treated as missing for scale analysis according to the locked data-processing procedure.

The AI-literacy analysis retained all five original AI-literacy items. Scale scores were calculated as the mean of valid items, with a minimum of 80% valid items required.

5.2 AI-Literacy Item-Level Findings

The five FITPED AI-literacy items showed substantial differences in their observed mean scores.

L1 had the highest observed mean (**4.559**), followed closely by L2 (**4.529**). L4 had the lowest observed mean (**3.230**). L3 and L5 occupied intermediate positions at **3.954** and **3.684**, respectively.

The results therefore show that perceived AI literacy was not uniform across the five items. The observed responses were higher for recognition of AI applications involving image and speech recognition than for the item concerning communication with an AI voice assistant.

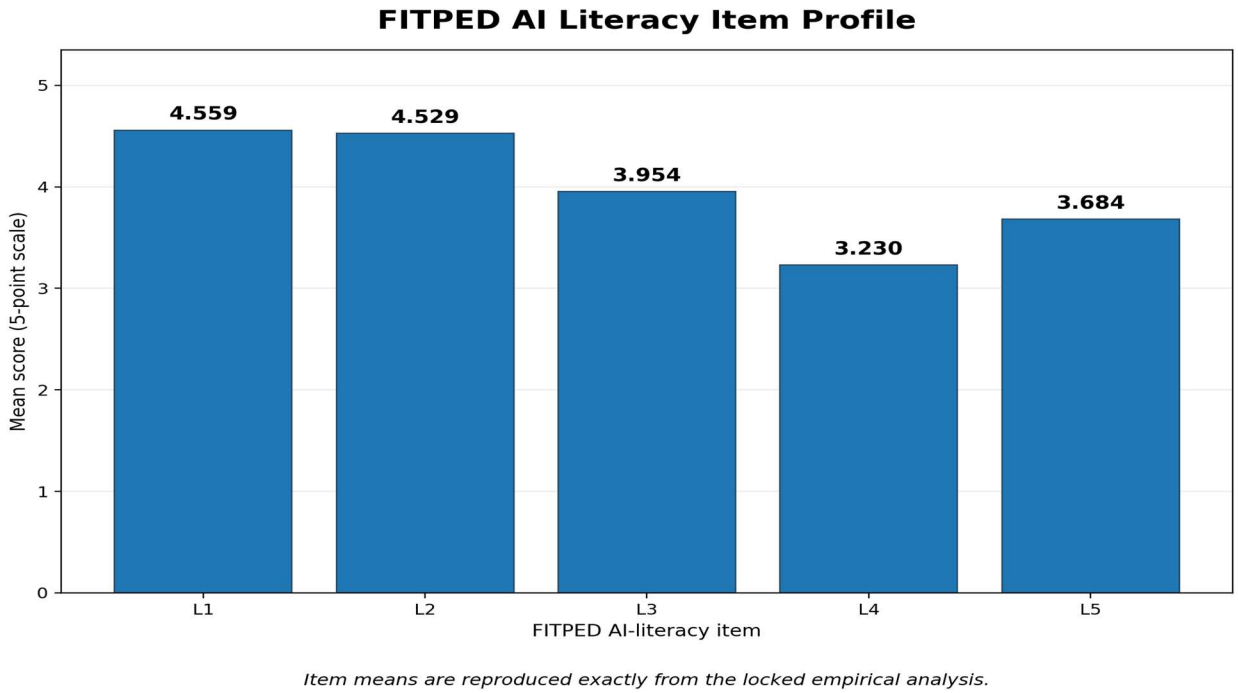


Figure 3. FITPED AI literacy item profile. Item means are shown exactly as reported in the locked empirical analysis (L1–L5).

5.3 AI-Literacy Scale and Reliability

The five-item AI-literacy scale produced the following descriptive statistics:

N = 1,205; Mean = 3.991; SD = 0.676; Median = 4.0; Cronbach's α = 0.688.

The scale therefore showed a relatively high observed mean within the five-point response framework, with moderate internal consistency.

The reliability analysis identified L4 as the item with the comparatively weaker item-total contribution. Removing L4 would increase Cronbach's alpha to approximately **0.743**. However, L4 was retained in the principal analysis because it is an original component of the questionnaire. The reported results therefore use the complete five-item scale rather than a post hoc reduced scale.

5.4 Other FITPED Construct Scores

The analysis also produced descriptive and reliability statistics for the related FITPED constructs.

5.5 Group Differences

5.5.1 Gender

Perceived AI-literacy scores differed across the observed gender groups.

- Male: **N = 793; Mean = 4.113**
- Female: **N = 402; Mean = 3.752**
- Other: **N = 10; Mean = 3.900**

The raw male–female difference was approximately **0.361 points**, with an unadjusted standardized difference of approximately **d = 0.55**.

5.5.2 Programme

The observed programme-level means were:

- IT: **N = 832; Mean = 4.107**
- Education: **N = 188; Mean = 3.730**
- Other: **N = 68; Mean = 3.794**
- Management: **N = 52; Mean = 3.796**
- Language: **N = 28; Mean = 3.471**
- IT education: **N = 22; Mean = 3.891**
- STEM education: **N = 14; Mean = 3.471**
- it: **N = 1; Mean = 4.400**

The raw IT–Education difference was approximately **0.377 points**, corresponding to an unadjusted standardized difference of approximately **d = 0.57**.

5.6 Year-Specific Results

The observed AI-literacy means by survey year were:

- **2022:** N = 535; Mean = 4.084
- **2023:** N = 112; Mean = 4.134
- **2024:** N = 558; Mean = 3.873

5.7 Matched 2022–2024 Subset

The repeated participant IDs permitted a within-person comparison for **59 participants** with observations in both 2022 and 2024.

The mean AI-literacy score was:

- 2022: **4.044**
- 2024: **3.898**

The mean within-person change was **−0.146 points**.

Among the 59 participants:

- **24 increased**
- **26 decreased**
- **9 remained unchanged**

The paired Wilcoxon analysis produced **p = .324**, with a rank-biserial effect of approximately **−0.159**.

Source: FITPED repeated-ID analysis; locked Step 5 results.

Note: The matched subset is small and should not be interpreted as a population-level longitudinal sample.

5.8 Correlations with AI Literacy

AI literacy was positively associated with all of the other reported FITPED constructs except AI anxiety.

The strongest association was with **AI readiness (ρ = .580)**. AI literacy was also positively associated with relevance (**ρ = .484**), career motivation (**ρ = .450**), behavioural intention (**ρ = .407**), social goods (**ρ = .367**), intrinsic motivation (**ρ = .365**), confidence (**ρ = .335**), and satisfaction (**ρ = .330**).

AI anxiety showed a modest negative association with AI literacy (**ρ = −.165**).

Source: FITPED correlation analysis; locked Step 5 results.

Note: These are associations and do not establish causal direction.

5.9 Exploratory Regression

An exploratory ordinary least squares model examined AI literacy as the dependent variable using year, age, gender, IT-programme status, and log-transformed AI-course hours as predictors. Standard errors were clustered by user_id.

The model produced **R^2 = .090**.

Source: FITPED exploratory OLS analysis; locked Step 5 results.

Note: Coefficients represent adjusted associations within the observed FITPED sample. They are not causal effects.

The 2024 coefficient was negative relative to 2022, whereas the 2023 coefficient was close to zero and statistically non-significant. Age showed a small positive adjusted association. The male and IT-programme indicators remained positively associated with AI literacy after adjustment.

AI-course hours had a small positive bivariate association with AI literacy (**Spearman ρ = .104, p < .001**), but the adjusted regression coefficient was **β = .005, p = .717**.

5.10 Regression Sensitivity

The principal sensitivity issue concerned the internal consistency of the five-item AI-literacy scale.

L4 made a comparatively weaker contribution to the scale. If L4 were removed, Cronbach's alpha would increase from **.688 to approximately .743**. Nevertheless, the item was retained because it is part of the original questionnaire.

The course-hours analysis also provided a sensitivity distinction between bivariate and adjusted relationships. Course hours were positively associated with AI literacy at the bivariate level (**ρ = .104, p < .001**), whereas the adjusted coefficient was not statistically significant (**β = .005, p = .717**).

The results therefore do not support treating course exposure as an independent predictor of perceived AI literacy within the fitted exploratory model.

5.11 Country-Level Descriptive Evidence

AI-literacy means varied across the countries represented in FITPED:

Source: FITPED country-level descriptive analysis; locked Step 5 results.

Note: Country categories are highly unequal in sample size. Country and year are also partly confounded. These figures are descriptive and should not be interpreted as national population estimates or causal country effects.

The country/year structure is particularly relevant because Indonesia and Lithuania are represented only in 2023, while the 2022 and 2024 observations are concentrated in Slovakia, Poland, and the Czech Republic. Consequently, apparent country differences cannot be separated cleanly from differences in year and sample composition.

5.12 Cross-Dataset Official Evidence

The FITPED results describe perceived AI literacy within a specific individual-level sample. Independent official datasets provide a different layer of evidence concerning AI use and structural digital inequality.

5.12.1 Eurostat: GenAI Use in the European Union

Eurostat's 2025 evidence for individuals aged 16–74 in the EU-27 reports:

The reported percentages represent proportions within the corresponding Eurostat population or subgroup/category. The overall 32.7% is the proportion of EU-27 individuals aged 16–74 using GenAI tools during the reference period, while each subgroup percentage is a proportion within its corresponding Eurostat subgroup/category. These denominators are distinct from the separate internet-user denominator used in other evidence streams.

- Overall GenAI use: **32.7%**
- Private use: **25.1%**
- Work use: **15.1%**
- Formal education use: **9.4%**

The age distribution shows a substantial difference:

- Ages 16–24: **63.8%**
- Ages 65–74: **8.0%**
- Difference: **55.8 percentage points**

By educational attainment:

- High formal education: **50%**
- Low/no formal education: **26%**
- Difference: **24 percentage points**

By location:

- Cities: **42%**
- Rural areas: **27%**
- Difference: **15 percentage points**

By gender:

- Men: **37%**
- Women: **33%**
- Difference: **4 percentage points**

By employment:

- Employed: **38%**
- Unemployed: **32%**

Professional use was:

- **22%** among employed people
- **11%** among unemployed people

Students reported GenAI use at **73%**, while economically inactive people reported **13%**. This difference is **60 percentage points**, although the categories differ substantially in age and life-stage composition.

Across EU countries, reported GenAI use ranged from:

- Denmark: **48.4%**
- Romania: **17.8%**

This represents a range of **30.6 percentage points**.

Eurostat's reported reasons for non-use included:

- **39%**: no necessity
- **8%**: lack of knowledge
- **5%**: did not know the tools existed
- **4%**: privacy/security concerns

These findings measure reported GenAI use and reasons for non-use. They do not constitute direct measures of demonstrated AI capability or realised benefit.

5.12.2 ITU: Digital Access in India

ITU data provide structural evidence concerning Internet access in India.

For 2025:

- Individuals using the Internet: **70.0%**
- Urban: **79.9%**
- Rural: **64.9%**
- Urban-rural difference: **15.0 percentage points**
- Male: **76.7%**
- Female: **63.1%**
- Male-female difference: **13.6 percentage points**

Households with Internet access at home were:

- Urban: **91.6%**
- Rural: **83.3%**
- Urban-rural difference: **8.3 percentage points**

For 2024, active mobile-broadband subscriptions were **61.9 per 100 inhabitants**, while at least 5G coverage was **82.2%**.

5.12.3 World Bank WDI

The World Bank WDI indicator IT.NET.USER.ZS reports **70.0%** of India's population using the Internet in 2025.

This corresponds to the ITU indicator and is therefore treated as corroborating evidence within the same international statistical system rather than as an independent survey estimate.

5.12.4 UNESCO: Digital Literacy

UNESCO's digital-literacy framework defines digital literacy in terms of confident and critical use of digital technologies for information, communication, and basic problem-solving.

Its SDG 4.4.2 framework concerns the percentage of youth and adults reaching minimum digital-literacy proficiency.

This evidence establishes an important measurement distinction: **digital literacy is broader than GenAI literacy**. UNESCO's framework therefore provides a capability context but does not supply a population-wide GenAI-literacy estimate for India within the assembled evidence.

5.13 Supporting International Evidence

PIAAC

PIAAC Cycle 2 Round 1 covers **31 countries/economies**, with data collection between **September 2022 and August 2023**. India was not included in Round 1.

Its relevance to the present results is methodological: PIAAC demonstrates the availability of large-scale performance-based adult skills measurement, but it does not measure GenAI literacy.

ICILS

ICILS 2023 covered **35 education systems** and more than **130,000 students**.

The international Computer and Information Literacy average was **476**, with approximately:

- **50%** of students reaching at least Level 2;
- **24%** below Level 1;
- **27%** at Level 1.

These figures concern Computer and Information Literacy rather than GenAI literacy.

Pew Research Center

The 2025 Pew evidence across 25 countries found a median of:

- **34%** who had heard or read a lot about AI;
- **47%** who had heard a little;
- **14%** who had heard nothing.

For India, **14%** reported having heard or read a lot about AI in the relevant survey.

This evidence measures AI awareness, not AI literacy or demonstrated capability.

The 2026 Pew survey included **42,151 adults across 36 countries**, with fieldwork from February 8 to May 13, 2026. Higher-education groups were more likely to report having heard a lot about AI in every country studied, while the median share reporting a lot of awareness was **50% in high-income countries** and **27% in middle-income countries**.

Stanford AI Index

The 2026 Stanford AI Index reports rapid diffusion of AI technologies, including a reported **53% population adoption within three years of launch**, **88% organisational AI adoption** in the reported measure, and roughly **four in five U.S. high-school and college students using AI for schoolwork**.

These figures provide adoption context but do not constitute measures of individual GenAI capability or realised benefit.

ILO

ILO analysis estimates that **25% of global employment** is potentially exposed to GenAI, rising to **34% in high-income countries**.

The exposure measure concerns occupational task exposure. It does not establish that exposed workers will lose employment, become more productive, possess stronger AI capability, or realise individual benefits.

World Bank/Semrush

The World Bank's 2025 GenAI-adoption analysis reports ChatGPT penetration among Internet users of:

- High-income countries: **24%**
- Upper-middle-income countries: **5.8%**
- Lower-middle-income countries: **4.7%**
- Low-income countries: **0.7%**

Average monthly visits to the top 60 GenAI tools per user were reported as:

- High-income: **1.9**
- Upper-middle-income: **0.5**
- Lower-middle-income: **0.4**
- Low-income: **0.08**

The high-income share of traffic to the top 360 GenAI tools increased from **55% to nearly 60%**, while the low-income share remained at approximately **0.2%**.

These are traffic/adoption proxies and should not be interpreted as individual-level measures of AI literacy, demonstrated capability, effective use, or benefit.

5.14 India and Global South Contextual Findings

The assembled evidence does not provide a representative individual-level estimate of AI literacy or demonstrated GenAI capability for India.

The strongest directly relevant India-wide evidence concerns **digital access**, not AI literacy. ITU reports Internet use at **70.0% in 2025**, with measurable urban-rural and gender differences.

Pew provides a separate measure of AI awareness, with **14% of Indian respondents** in the relevant 2025 survey reporting that they had heard or read a lot about AI.

Selected Indian higher-education studies provide evidence concerning ChatGPT adoption among specific student populations, including students in Punjab and northern Indian states. These studies are contextual rather than representative estimates of the Indian population.

The evidence from other Global South contexts, including Sri Lanka, similarly provides selected higher-education evidence concerning barriers and opportunities for GenAI adoption rather than a harmonised Global South estimate of AI literacy or capability.

Accordingly, the evidence supports the presence of **structural digital-access inequality and selected AI-awareness/adoption evidence in India and developing-country contexts**, but does not support an individual-level national estimate of AI literacy or demonstrated GenAI capability.

5.15 Measurement and Data-Gap Findings

The cross-dataset evidence revealed several stages for which measurement remains incomplete or insufficiently harmonised.

Demonstrated GenAI capability

Objective AI and GenAI capability measurement is now emerging. GLAT and AICOS demonstrate that performance-based assessment is feasible.

However, these measures are not yet integrated into the assembled population-level evidence architecture in a manner that simultaneously captures representative access, effective use, and realised benefit across the same individuals.

Critical AI-output evaluation

AI-literacy research includes evaluation and critical-appraisal dimensions, and selected GenAI-literacy instruments incorporate content evaluation. However, a harmonised population-level measure of the ability to critically evaluate GenAI outputs is not available across the assembled access and use datasets.

Effective GenAI use

Official use statistics establish whether people report using GenAI but generally do not establish whether that use is competent, critical, appropriate, or effective.

Consequently, effective use remains insufficiently harmonised as a population-level measure.

Individual realised GenAI benefit

Selected studies examine outcomes such as job performance or information-verification behaviour. However, these are study-specific outcomes in selected populations and cannot be combined into a common population-level measure of realised GenAI benefit.

Complete individual-level pathway

No assembled dataset simultaneously provides sufficiently harmonised individual-level observations for:

Access → Capability → Effective Use → Benefit.

The evidence therefore supports examination of separate stages and their conceptual relationships, but not estimation of the complete individual-level pathway.

Table 2. FITPED AI Literacy Descriptive and Reliability Results

Measure	Value
Number of records	1,205
Unique participants	1,146
AI-literacy items	5
Mean	3.991
SD	0.676
Median	4.0
Cronbach's α	0.688
α if L4 removed	≈ 0.743
L1 mean	4.559
L2 mean	4.529
L3 mean	3.954
L4 mean	3.230

L5 mean	3.684

Source: FITPED AI Literacy Questionnaire dataset; locked Step 5 analysis.

Note: L4 was retained because it is an original questionnaire item; α if item deleted is reported only as a sensitivity statistic.

Table 3. FITPED Associations and Exploratory Regression

Analysis	Variable / Predictor	Estimate	95% CI	p
Correlation (Spearman ρ)	AI Readiness	.580	—	—
Correlation (Spearman ρ)	Relevance	.484	—	—
Correlation (Spearman ρ)	Career Motivation	.450	—	—
Correlation (Spearman ρ)	Behavioural Intention	.407	—	—
Correlation (Spearman ρ)	Social Goods	.367	—	—
Correlation (Spearman ρ)	Intrinsic Motivation	.365	—	—
Correlation (Spearman ρ)	Confidence	.335	—	—
Correlation (Spearman ρ)	Satisfaction	.330	—	—
Correlation (Spearman ρ)	AI Anxiety	−.165	—	—
Regression	2023 vs. 2022	.018	[−.141, .176]	.827
Regression	2024 vs. 2022	−.108	[−.185, −.032]	.006
Regression	Age	.007	[.000, .015]	.043
Regression	Male vs. female	.217	[.120, .315]	< .001
Regression	IT programme	.210	[.106, .313]	< .001
Regression	log(1 + AI-course hours)	.005	[−.020, .029]	.717
Regression	Model R^2	.090	—	—

Source: FITPED Step 5 analysis.

Note: Correlation coefficients are associations only. Regression coefficients are adjusted associations from the exploratory OLS model with cluster-robust standard errors at participant level; they do not establish causality.

Table 4. Eurostat GenAI-Use Inequalities, 2025

Dimension	Comparison	Observed values

Overall use	EU-27, individuals aged 16–74	32.7%
Age	16–24 vs. 65–74	63.8% vs. 8.0%
Education	High vs. low/no formal education	50% vs. 26%
Location	Cities vs. rural	42% vs. 27%
Gender	Men vs. women	37% vs. 33%
Employment	Employed vs. unemployed	38% vs. 32%
Professional use	Employed vs. unemployed	22% vs. 11%
Students	Students	73%
Economically inactive	Economically inactive	13%
Country	Denmark vs. Romania	48.4% vs. 17.8%

Source: Eurostat dataset isoc_ai_iaiu; 2025; EU-27; individuals aged 16–74; DOI 10.2908/isoc_ai_iaiu.

Note: These are population-level GenAI-use indicators, not direct measures of demonstrated AI capability or realised individual benefit.

Table 5. India Digital-Access Inequalities

Indicator	Year	Value
Individuals using the Internet	2025	70.0%
Urban Internet use	2025	79.9%
Rural Internet use	2025	64.9%
Urban-rural gap	2025	15.0 pp
Male Internet use	2025	76.7%
Female Internet use	2025	63.1%
Male-female gap	2025	13.6 pp
Urban households with Internet at home	2025	91.6%
Rural households with Internet at home	2025	83.3%
Urban-rural household gap	2025	8.3 pp
Active mobile-broadband subscriptions	2024	61.9 per 100
At least 5G coverage	2024	82.2%

Source: ITU DataHub; corroborated by World Bank WDI indicator IT.NET.USER.ZS.

Note: These indicators measure digital access/use conditions. They are not AI-literacy or demonstrated GenAI-capability measures. WDI IT.NET.USER.ZS provides corroborating evidence for the 70.0% Internet-use value.

6. Cross-Dataset Evidence Synthesis

The combined evidence indicates that the emerging AI literacy divide is better understood as a multidimensional inequality than as a single difference in access to AI technologies. The FITPED analysis provides direct individual-level evidence concerning perceived AI literacy, while the independent official and international datasets provide evidence concerning GenAI use, digital access, broader digital capability, AI awareness, adoption, and occupational exposure. These evidence streams converge on the importance of distinguishing stages of AI participation, but they do not constitute one pooled empirical population.

The central measurement architecture can therefore be represented as:

Digital Divide → AI Access Divide → AI Literacy Divide → Capability Divide → Effective-Use Divide → Benefit Divide

The available evidence supports observation of several of these stages independently. It does not support estimating the complete sequence within the same individuals.

At the first stage, the broader digital divide remains relevant. ITU and World Bank WDI data show that Internet access in India reached 70.0% in 2025, while substantial urban-rural and gender differences remained. These data demonstrate that access conditions continue to differ within populations. They do not, however, measure AI literacy or GenAI capability.

At the AI-use stage, Eurostat provides evidence of substantial differences in reported GenAI use across age, education, location, employment, and country. The difference between 16–24-year-olds and those aged 65–74, for example, was substantial in the 2025 EU-27 data. Yet these figures measure use rather than demonstrated competence. They therefore establish an AI-use divide without establishing an equivalent capability divide.

The FITPED findings occupy a different position. They provide direct individual-level evidence of perceived AI literacy within a student-oriented sample. The overall mean of 3.991 indicates relatively high perceived literacy within the observed five-point scale, but the item means demonstrate that perceived competence is not uniform across different AI applications. The strongest item means concerned image and speech recognition, while communication with an AI voice assistant received the lowest mean.

The FITPED associations further demonstrate that perceived AI literacy is connected to several related perceptions and intentions. AI readiness showed the strongest association with AI literacy ($\rho = .580$), followed by relevance ($\rho = .484$), career motivation ($\rho = .450$), and behavioural intention ($\rho = .407$). AI anxiety was negatively associated with perceived AI literacy ($\rho = -.165$). These associations establish relationships within the observed sample, but they do not determine causal direction.

The broader evidence therefore suggests an important distinction: **access and use can be measured at population scale more readily than demonstrated AI capability, effective use, and realised benefit**.

This distinction is central to understanding the AI literacy divide.

Table 6. Cross-Dataset Measurement Framework

Construct	Operational meaning in this study	Empirical status
Perceived AI literacy	Self-reported AI-literacy score based on FITPED L1–L5	Directly measured
GenAI use	Reported or population-level use of generative AI systems	Measured through independent aggregate evidence

Digital access	Internet/digital connectivity and access conditions	Measured through ITU/WDI and related official evidence
Digital capability	Broader demonstrated digital/computer-information competencies	Contextually measured through PIAAC/ICILS/UNESCO evidence
Demonstrated GenAI capability	Performance-based ability to complete or demonstrate GenAI-related competencies	Emerging evidence; not directly measured in FITPED
Effective GenAI use	Competent, appropriate, critical, or productive application of GenAI	Insufficiently harmonised in available population data; data gap
Realised individual GenAI benefit	Individual educational, occupational, economic, informational, or social outcomes attributable to effective AI use	Insufficiently harmonised individual-level evidence; data gap
Educational inequality	Differences associated with educational attainment/programme context	Partly measured through FITPED programme groups and independent aggregate evidence
Socioeconomic context	Structural economic or socioeconomic environment associated with access/adoption	Contextually represented through independent aggregate evidence; not a single FITPED SES variable

Source: Approved operationalisation matrix from the source manuscript.

Note: The table preserves construct distinctions and does not imply that the constructs form a validated causal sequence.

Table 7. Measurement-Gap Matrix

Framework stage	**Evidence available**	**Measurement status in present study**
Digital access	ITU/WDI	Direct aggregate evidence
AI access/adoption	Eurostat; World Bank/Semrush and related evidence	Aggregate/contextual evidence
Perceived AI literacy	FITPED	Direct individual-level evidence
AI awareness	Pew	Population survey evidence
Broader digital capability	UNESCO/PIAAC/ICILS	Contextual performance-based evidence
Demonstrated GenAI capability	GLAT/AICOS and emerging instruments	Emerging; not directly measured in FITPED
Effective GenAI use	Fragmented evidence	Insufficiently harmonised
Realised individual benefit	Selected outcome studies	Insufficiently harmonised

Complete Access → Capability → Effective Use → Benefit pathway	No integrated source	Not estimable

Source: Approved cross-dataset evidence architecture.

Note: The evidence supports examination of separate stages and their conceptual relationships, but not estimation of the complete individual-level pathway.

AI Inequality Evidence Map

Evidence source	Population / unit	Year	Measurement level	Primary construct
FITPED	Students; 1,146 unique participants	2022–2024	Individual / microdata	Perceived AI literacy; related constructs
Eurostat	EU-27 individuals aged 16–74	2025	Population aggregate	GenAI use
ITU DataHub	India: individuals / households	2025 access; 2024 mobile/5G	Population / household aggregate	Internet access and structural digital access
World Bank WDI	India population	2025	Population aggregate	Individuals using Internet
UNESCO UIS	Global framework / youth & adults	2018 framework; consulted 2025	Framework / population-level construct	Digital literacy / proficiency concept
OECD PIAAC	Adults; 31 countries/ economies, Round 1	2022–2023 data collection	Individual assessment / survey	Digital/problem-solving skills
IEA ICILS	Students; 35 education systems	2023	Individual assessment / surve	Computer and information literacy
Pew Research Center	Adults; 25 countries (2025 evidence	2025	Population survey aggregate	AI awareness
Stanford AI Index	Population; organizations; students (reported measures)	2026	Mixed aggregate / contextual	AI adoption context
ILO	Global employment / occupations	2025	Occupational aggregate	Potential GenAI exposure
World Bank / Semrush	Internet users by World Bank income group	April 2025	Traffic/adoption proxy aggregate	Consumer-facing GenAI adoption proxy

Evidence streams are not pooled at respondent level. Years and measurement levels are retained as reported; constructs are not treated as interchangeable.

Figure 4. AI inequality evidence map. The map preserves the population, year, measurement level and construct of each locked evidence source without treating them as interchangeable measures.

7. Discussion

7.1 From Digital Inequality to the AI Literacy Divide

The findings extend an established trajectory in digital-inequality research. Earlier digital-divide research distinguished access from skills, and later research extended the framework toward uses and outcomes (Hargittai, 2002; Robinson et al., 2015; Scheerder et al., 2017). The present evidence suggests that a similar differentiation is necessary in the AI domain.

The emergence of GenAI does not eliminate the traditional digital divide. Rather, it introduces additional questions about what individuals can do once digital and AI technologies become available. A person may have Internet access without having access to a particular AI service; may have access to an AI service without understanding its capabilities; may understand an AI system without being able to demonstrate effective use; and may use an AI system without obtaining a measurable benefit.

The evidence assembled in this study does not demonstrate that these stages necessarily occur sequentially for every individual. Instead, it shows that they represent analytically distinct forms of inequality that are currently measured through different evidence systems.

This interpretation is consistent with the broader digital-divide literature, which has repeatedly shown that access, skills, use, and outcomes should not be collapsed into one indicator of digital inclusion (Robinson et al., 2015; Scheerder et al., 2017).

7.2 What the FITPED Analysis Reveals About Perceived AI Literacy

The FITPED analysis provides the study's principal direct individual-level evidence.

The overall AI-literacy mean was **3.991**, with a standard deviation of **0.676** and median of **4.0** on the five-point scale. This indicates relatively high perceived AI literacy within the observed sample. However, the item-level results show meaningful variation across the five components.

L1 and L2 had means of **4.559** and **4.529**, respectively, whereas L4 had a mean of **3.230**. This pattern indicates that respondents did not express equivalent levels of perceived familiarity or intended engagement across all AI applications represented by the instrument.

The finding is important because it cautions against treating an overall AI-literacy score as a complete representation of AI competence. Even within one instrument, different AI-related capabilities or applications can receive different levels of endorsement.

The moderate internal consistency of the five-item scale ($\alpha = .688$) also supports a cautious interpretation. The instrument captures a coherent but not perfectly homogeneous construct. The fact that alpha would increase to approximately .743 if L4 were removed does not justify deleting the item, because doing so would modify the original measurement structure. Instead, the result illustrates the multidimensional character of the construct.

7.3 Demographic and Educational Differences

The FITPED data show differences in perceived AI literacy across observed demographic and educational groups.

Male participants had a mean AI-literacy score of **4.113**, compared with **3.752** among female participants. The raw difference was approximately **0.361**, with an unadjusted standardized difference of approximately **d = .55**.

Participants in IT programmes had a mean of **4.107**, compared with **3.730** among participants in Education programmes. The raw difference was approximately **0.377**, with an unadjusted standardized difference of approximately **d = .57**.

These differences remained observable in the exploratory regression, where the adjusted coefficient for male participants relative to female participants was **β = .217** and the adjusted coefficient for the IT-programme indicator was **β = .210**.

These results establish associations within the observed FITPED sample. They do not establish that gender or programme choice causes differences in AI literacy.

Several explanations remain compatible with the observed pattern. Differences could reflect educational exposure, prior technological experience, programme composition, self-selection into fields of study, access to AI-related learning opportunities, or other unmeasured characteristics. The available data cannot distinguish among these explanations.

This is particularly important because the FITPED sample is heavily represented by IT-related programmes. The observed programme difference therefore should not be interpreted as a general population estimate of the relationship between academic discipline and AI literacy.

7.4 Age, Year and the Problem of Temporal Interpretation

Age showed a small positive adjusted association with perceived AI literacy in the exploratory regression (**β = .007**, 95% CI [.000, .015], p = .043). However, this result should be interpreted within the characteristics of the sample and model rather than as evidence that age itself determines AI literacy.

The year-specific descriptive results were:

- 2022: **M = 4.084**
- 2023: **M = 4.134**
- 2024: **M = 3.873**

The exploratory regression also produced a negative 2024 coefficient relative to 2022 (**β = −.108**, 95% CI [−.185, −.032], p = .006).

However, the year coefficient cannot be interpreted as evidence of a population-wide decline in AI literacy. The composition of the FITPED sample differs across years, and country and year are partly confounded.

The matched analysis provides a more direct but much smaller longitudinal comparison. Among the 59 participants observed in both 2022 and 2024, the mean changed from **4.044 to 3.898**, corresponding to a mean within-person change of **−.146**. The paired Wilcoxon test was not statistically significant (**p = .324**).

The combined evidence therefore does not establish a systematic within-person temporal change. The difference between the cross-sectional year coefficient and the matched-subset result illustrates why changing sample composition must be considered when interpreting repeated cross-sectional observations.

7.5 Access Is Not Capability

The distinction between access and capability is one of the most important implications of the evidence.

India's 2025 Internet-use rate of **70.0%** indicates substantial digital connectivity, but this statistic cannot establish how many individuals possess AI literacy or demonstrated GenAI capability. The same applies to urban-rural and gender differences in Internet access. These figures establish unequal access conditions, not unequal AI competence.

Similarly, Eurostat's 2025 GenAI-use statistics establish differences in participation. For example, GenAI use was **63.8% among people aged 16–24**, compared with **8.0% among people aged 65–74**. The evidence establishes an age-related difference in reported use. It does not establish that the younger group possesses proportionately greater demonstrated capability or obtains greater benefit.

This distinction reflects the progression identified in the digital-divide literature. Hargittai's second-level digital divide showed that people who are already online can differ in skills, while Scheerder et al. demonstrated that skills, uses, and outcomes constitute analytically distinct levels of digital inequality.

The same principle should therefore be applied to AI. Increasing access can create the conditions for participation without establishing equivalent capability.

7.6 Perceived Literacy Is Not Demonstrated Capability

The distinction between perceived and demonstrated capability is particularly important because FITPED's principal construct is self-reported.

The FITPED results show what participants reported about their AI literacy. They do not establish whether participants could successfully complete an objective AI-related task.

This distinction is supported by the broader measurement literature. Chiu et al. (2024) developed an objective AI-literacy assessment specifically in response to limitations associated with self-reported measurement. GLAT subsequently extended this approach into GenAI-specific performance assessment. Jin et al. (2025) found that performance-based GenAI literacy was associated with GenAI-supported task performance more directly than self-reported proficiency in the study population.

The appropriate interpretation is therefore not that self-reported AI literacy is invalid. Rather, perceived literacy and demonstrated capability answer different empirical questions.

This distinction materially affects interpretation of the FITPED mean of 3.991. The result should be described as relatively high **perceived AI literacy within the sample**, not as evidence that 1,205 observations represent a population with equivalent demonstrated GenAI competence.

7.7 Objective GenAI Capability Is Emerging but Not Yet Integrated

The evidence base has moved beyond purely self-reported assessment.

GLAT provides a performance-based GenAI literacy assessment, while AICOS represents another objective approach to AI competency that incorporates GenAI literacy. These instruments demonstrate that objective assessment is increasingly feasible.

However, their existence does not resolve the broader measurement-integration problem.

Objective GenAI-capability measurement is **emerging, but remains insufficiently integrated with representative measures of access, effective use and realised benefit across the same individuals and comparable populations**.

This is a more precise interpretation of the evidence than either of two extremes. It would be incorrect to say that objective capability cannot be measured. It would also be incorrect to treat the existence of several performance-based instruments as evidence that a complete population-level measurement system already exists.

The current evidence therefore provides a partial solution to the capability-measurement problem while leaving the broader integration problem unresolved.

7.8 From Capability to Effective Use

Demonstrated capability and actual use are not necessarily equivalent.

A person may demonstrate competence under an assessment condition without using AI frequently in everyday life. Conversely, a person may use AI frequently without demonstrating strong ability to evaluate outputs or apply AI appropriately.

This distinction is particularly relevant to GenAI because use can encompass very different tasks and levels of complexity.

Eurostat provides population-level evidence concerning whether people use GenAI and for what broad contexts, but these measures do not establish the quality of interaction. Similarly, World Bank/Semrush evidence provides traffic and adoption proxies but does not establish effective individual use.

Selected research begins to address this problem by linking GenAI literacy to specific behaviours. Yan et al. (2026) examined relationships with privacy-protection and information-verification behaviours, while Liu et al. (2025) examined the relationship between GenAI literacy and job performance. These studies demonstrate that researchers can investigate consequences of GenAI literacy, but they do not provide a harmonised population-wide measure of effective use.

The distinction therefore remains a central unresolved stage of the proposed framework.

7.9 Effective Use and Realised Benefit

The final stages of the conceptual architecture concern what people actually gain from AI use.

The existing evidence provides examples of outcome measurement, but these outcomes are context-specific. Job performance, information verification, privacy behaviour, educational performance, and other possible benefits represent different constructs and cannot automatically be treated as a common measure of realised GenAI benefit.

The World Bank/Semrush adoption evidence further illustrates the distinction. Higher-income economies show substantially greater consumer-facing GenAI engagement than lower-income economies in the reported traffic

measures. This establishes an adoption inequality but cannot establish that people in higher-income economies necessarily obtain proportionately greater individual benefits from AI.

The ILO's occupational-exposure estimates provide another contextual layer. Potential exposure to GenAI is not equivalent to actual use, capability, productivity improvement, employment loss, or individual benefit.

Consequently, the available evidence does not permit the final stage of the conceptual pathway to be estimated consistently. The benefit divide remains substantially less directly observed than the access and adoption dimensions.

7.10 Educational and Socioeconomic Inequality

7.11 Alternative Explanations and Interpretive Boundaries

The combined evidence is consistent with the proposition that educational and socioeconomic context matters for AI participation, but it does not establish a simple deterministic relationship.

Eurostat shows a difference in GenAI use between individuals with high formal education (**50%**) and those with low/no formal education (**26%**). It also shows substantial differences by employment and location.

The World Bank/Semrush evidence indicates pronounced differences in consumer-facing GenAI adoption across income groups, with reported ChatGPT penetration among Internet users of **24% in high-income countries**, compared with **0.7% in low-income countries**.

These findings demonstrate that AI participation occurs within unequal social and economic environments. They do not demonstrate that socioeconomic status directly causes AI capability or benefit.

The distinction is important because socioeconomic context can affect multiple upstream and downstream conditions simultaneously: connectivity, device availability, educational opportunity, workplace exposure, language resources, time, institutional support, and access to AI-enabled services. The available cross-sectional aggregate evidence cannot isolate these mechanisms.

7.12 Geography and Country Differences

Geographic differences appear at several levels of the evidence architecture.

Within Eurostat, cities reported higher GenAI use than rural areas (**42% versus 27%**). India likewise shows an urban-rural Internet-use difference of **15.0 percentage points**.

The FITPED data show differences among represented countries, but those comparisons are strongly affected by unequal sample sizes and country/year confounding. They therefore should not be interpreted as national AI-literacy rankings.

The cross-country evidence illustrates why geographical comparisons require particular caution. A difference between two countries may reflect digital infrastructure, education systems, economic structure, age composition, survey design, institutional exposure, or other contextual factors.

7.13 Practical and Research Implications

The findings suggest several implications for measurement and research infrastructure.

First, future AI-literacy surveys should clearly distinguish self-perceived competence from demonstrated capability. A single self-report score should not be treated as a substitute for performance-based assessment.

Second, large-scale population studies could benefit from combining access indicators with validated AI/GenAI capability assessments. This would allow researchers to examine whether differences in access correspond to differences in demonstrated competence rather than assuming such a relationship.

Third, measures of AI use should be complemented where possible by indicators of effective use. Frequency of interaction alone provides limited information about whether AI was used appropriately or whether its outputs were critically evaluated.

Fourth, future studies should develop comparable outcome measures for realised individual benefit. Educational, occupational, informational, and other outcomes should be distinguished rather than collapsed into a generic concept of "AI benefit."

Fifth, longitudinal designs are needed to observe whether individuals move between stages of AI participation over time. The FITPED matched subset illustrates both the potential value and the current limitation of such evidence: only 59 participants could be followed directly across 2022 and 2024.

Sixth, broader representation is needed. Future measurement systems should include diverse educational levels, occupational groups, age groups, geographic settings, and socioeconomic contexts rather than relying predominantly on selected student populations.

These are research and measurement implications derived from the observed evidence architecture. They are not claims that any particular intervention has already been demonstrated to produce a specific outcome.

8. Theoretical and Methodological Contributions

8.1 Theoretical Contribution

The primary theoretical contribution of this study is the development of an integrated measurement framework for understanding the emerging AI literacy divide. Rather than treating AI inequality as a single problem of access to artificial intelligence, the framework organises the evidence across six analytically distinct stages:

Digital Divide → AI Access Divide → AI Literacy Divide → Capability Divide → Effective-Use Divide → Benefit Divide

This architecture extends the logic of established digital-divide research into the context of generative AI while preserving distinctions between access, knowledge and literacy, demonstrated capability, patterns of use, and outcomes. The framework therefore provides a structure through which apparently similar forms of AI inequality can be distinguished and empirically examined.

The framework should be understood as a **conceptual and measurement architecture rather than a newly demonstrated causal pathway**. The present study does not establish that access necessarily produces AI literacy, that AI literacy necessarily produces demonstrated capability, or that capability necessarily produces effective use and realised benefit. Instead, it identifies these dimensions as analytically connected stages that require more integrated empirical measurement.

This distinction is particularly important because the available evidence shows that different datasets measure different parts of the architecture. Internet access indicators can document connectivity without measuring AI capability; AI-use indicators can document whether people use GenAI without establishing whether that use is effective; and self-reported AI-literacy measures can capture perceived understanding without demonstrating task-level competence. The framework consequently provides a basis for interpreting these measures without treating them as interchangeable.

The contribution is therefore not a new universal estimate of AI literacy. It is a structured account of **what needs to be measured to understand AI inequality comprehensively**.

8.2 Methodological Contribution

The methodological contribution lies in combining individual-level evidence from the FITPED dataset with independent population-level, international, and contextual evidence while maintaining the distinct evidentiary status of each source.

FITPED provides the study's primary individual-level empirical evidence concerning perceived AI literacy and its relationships with related AI attitudes, readiness, motivation, confidence, anxiety, satisfaction, and behavioural intention. Independent datasets provide complementary evidence concerning GenAI use, digital access, AI awareness, digital capability, and broader AI adoption. These sources are therefore used as complementary evidence rather than being treated as observations from a single pooled population.

A central methodological principle is the preservation of **measurement-level distinctions**. The study does not treat digital access as equivalent to AI literacy, perceived AI literacy as equivalent to demonstrated AI capability, AI use as equivalent to effective use, or educational attainment as a direct measure of AI capability. This prevents conceptually different indicators from being collapsed into a single construct merely because they are related to AI inequality.

The analysis also avoids inappropriate respondent-level merging of datasets. FITPED observations are not combined with Eurostat, ITU, UNESCO, PIAAC, ICILS, Pew, World Bank, Stanford, or ILO observations as though they represented the same individuals. Instead, the datasets are interpreted according to their respective populations, measurement designs, and analytical levels.

A further methodological contribution is the explicit separation of **perceived literacy from demonstrated capability**. The FITPED AI Literacy construct is retained as a measure of perceived/recognised AI literacy consistent with its questionnaire structure. At the same time, the literature review establishes that objective and performance-based GenAI-capability measures are emerging. The study therefore avoids interpreting self-reported literacy as direct evidence of task-level competence.

Similarly, the study distinguishes **use from effective use**. A person may use a generative AI system without the available evidence establishing whether the resulting use is accurate, critical, efficient, appropriate, or beneficial. Consequently, frequency or prevalence of use is not interpreted as evidence of effective AI use.

Finally, the study explicitly documents the absence or fragmentation of outcome measures. Realised individual benefit cannot be inferred simply from access, awareness, literacy, or use. This makes the missing outcome dimension an empirical limitation of the evidence ecosystem rather than a gap that can be filled through assumption.

8.3 Measurement/Data Gap

The principal measurement gap identified by the study is the insufficient integration of five dimensions:

access → perceived literacy → demonstrated GenAI capability → effective use → realised individual benefit.

Current evidence provides substantial information about individual components of this sequence, but these components are distributed across different datasets, populations, instruments, and levels of measurement.

Access can be measured through Internet connectivity and related infrastructure indicators. GenAI-use datasets can measure whether and how frequently people report using generative AI. AI-literacy studies can measure perceived knowledge, attitudes, competencies, or performance. Emerging objective assessments can evaluate demonstrated AI or GenAI capability. However, these dimensions are rarely measured together across the same individuals using harmonised instruments, particularly alongside measures of effective use and realised individual benefit.

The critical gap is therefore not the complete absence of objective GenAI-capability measurement. **Objective GenAI-capability measurement is emerging, but it remains insufficiently integrated with representative measures of access, effective use, and realised benefit across the same individuals and comparable populations.**

This distinction is essential. The evidence base is developing beyond purely self-reported measures, but the development of individual capability assessments has not yet resolved the broader integration problem identified by this study.

The resulting measurement discontinuity limits the extent to which researchers can determine how inequalities in access and perceived literacy relate to demonstrated competence, how competence relates to effective use, and whether effective use translates into meaningful individual outcomes.

The study therefore identifies a **measurement-integration gap**, rather than claiming that any single component of AI literacy research is absent.

9. Limitations and Future Research

9.1 Limitations of the Present Study

The first limitation concerns the FITPED sample. FITPED is not a population-representative sample of all AI users or the general population. It is a student-oriented dataset with substantial representation from IT-related programmes and therefore should not be interpreted as providing a universal estimate of AI literacy.

Second, important FITPED constructs are based on self-report. The AI Literacy measure captures participants' responses to the questionnaire items and therefore should not be interpreted as a direct performance-based assessment of demonstrated GenAI capability. This distinction is central to the interpretation of the study's findings.

Third, the longitudinal component is limited. Although the dataset contains repeated identifiers and includes observations from 2022 and 2024 for a subset of participants, only 59 participants could be identified as having matched observations across those years. The matched analysis therefore provides limited longitudinal evidence and cannot support broad population-level conclusions about change over time.

Fourth, country and year are not fully separable within the FITPED evidence. Some countries are represented primarily in particular years, while the 2022–2024 observations are concentrated in particular country groups. Consequently, observed country or year differences cannot be interpreted as clean country effects or causal temporal trends.

Fifth, the study combines evidence produced by different populations, sampling approaches, instruments, definitions, and collection periods. Cross-dataset comparisons are therefore descriptive and analytical rather than equivalent statistical comparisons between common populations.

Sixth, the study does not construct a unified composite AI-inequality index. Such an index would risk combining measures operating at different conceptual and observational levels. Preserving these distinctions is methodologically preferable for the present research design.

9.2 Limitations of the Available Evidence Ecosystem

The broader evidence ecosystem presents additional limitations that extend beyond the design of the present study.

First, representative individual-level evidence simultaneously measuring access, perceived AI literacy, demonstrated GenAI capability, effective use, and realised benefit remains limited. Existing datasets often capture only one or several of these dimensions.

Second, effective use remains incompletely measured at broad population level. Evidence on whether AI is used effectively, critically, accurately, and appropriately is substantially less integrated than evidence on whether AI is accessed or used.

Third, realised individual benefit is similarly incompletely measured. Current evidence does not provide a sufficiently integrated basis for determining whether differences in AI access, literacy, capability, or use translate into comparable differences in individual outcomes.

Fourth, representative India-specific evidence on AI literacy remains limited. The available evidence supports important observations about Internet access, AI awareness, and selected studies of AI adoption, but it does not provide a representative national estimate of AI literacy or demonstrated GenAI capability.

Fifth, international evidence uses different constructs and instruments. Digital literacy, computer and information literacy, AI literacy, GenAI literacy, AI awareness, Internet use, and GenAI adoption cannot be assumed to measure the same underlying construct.

Finally, much of the available evidence operates at either the individual or aggregate level, creating a fundamental limitation for attempts to connect structural conditions with individual capabilities and outcomes.

These limitations do not invalidate the evidence used in the study. Rather, they define the boundaries within which the evidence can be interpreted.

9.3 Future Research

Future research should move from fragmented measurement toward longitudinal, individual-level designs capable of observing multiple stages of the proposed architecture within the same participants.

A particularly important empirical objective is to measure, within the same individuals:

Access → AI literacy → demonstrated capability → effective use → realised benefit.

Such designs would make it possible to determine whether differences observed at one stage correspond to differences at subsequent stages without assuming the relationships in advance. The present framework should therefore be treated as a research architecture for future empirical testing rather than as an already validated causal model.

Future studies should incorporate objective and performance-based assessments of GenAI capability alongside self-reported literacy measures. Existing developments in objective AI-literacy and performance-based GenAI-literacy assessment provide a basis for moving beyond exclusive reliance on self-perception.

Measurement should also extend beyond whether AI is used to examine **how effectively it is used**. Future instruments could therefore operationalise dimensions such as the evaluation of AI outputs, appropriate verification, critical assessment, and task-level performance where these can be empirically measured. These are future measurement requirements, not findings established by the present study.

Another priority is the development of harmonised effective-use measures that can be applied across populations and countries. Comparable instruments would allow researchers to distinguish simple AI adoption from competent and effective utilisation.

Representative datasets are particularly important for India and other underrepresented populations. Future studies should seek population-level evidence that simultaneously captures access, AI literacy, demonstrated capability, use, effective use, and outcomes rather than relying on selected educational or occupational samples alone.

Comparable cross-country instruments are also necessary for determining whether observed differences reflect genuine inequalities or differences in measurement design. Harmonisation of definitions, sampling procedures, assessment formats, and outcome measures would substantially improve the interpretability of international comparisons.

Finally, longitudinal designs should measure realised individual outcomes rather than stopping at access or adoption. This would allow future research to investigate whether demonstrated capability and effective use are

associated with meaningful benefits and whether inequalities persist, narrow, or widen across successive stages of AI engagement.

All of these represent **future research requirements**, not conclusions demonstrated by the present study.

10. Conclusion

This study examined the emerging AI literacy divide by bringing together individual-level evidence from FITPED with independent evidence on GenAI use, digital access, AI awareness, digital capability, and broader AI adoption. The evidence demonstrates that AI inequality is multidimensional. Differences are observable not only in digital access and GenAI use, but also in perceived AI literacy, AI awareness, and related patterns of capability-relevant evidence.

The study's primary contribution is an integrated measurement framework that distinguishes six analytically important stages:

Digital Divide → AI Access Divide → AI Literacy Divide → Capability Divide → Effective-Use Divide → Benefit Divide.

The framework does not claim that these stages constitute an empirically demonstrated causal sequence. Instead, it provides a structure for understanding why evidence from access, literacy, capability, use, and outcomes should not be treated as interchangeable.

The study also establishes an important measurement boundary. Objective GenAI-capability assessment is emerging, but evidence remains insufficiently integrated with representative measures of access, effective use, and realised individual benefit across the same individuals and comparable populations. Consequently, the complete transition from access and literacy to capability, effective use, and benefit remains unresolved.

This unresolved measurement problem is central to understanding the AI literacy divide. If access is measured without capability, inequality may be underestimated. If self-reported literacy is treated as demonstrated competence, capability differences may be obscured. If use is treated as effective use, adoption may be mistaken for meaningful engagement. And if outcomes are not measured, it remains impossible to determine whether differences in AI engagement translate into differences in realised individual benefit.

The central empirical insight is therefore that **AI inequality is increasingly multidimensional, while the evidence used to measure it remains fragmented across access, literacy, capability, use, and outcomes**.

Understanding the AI literacy divide consequently requires not only asking who has access to AI or who uses it, but developing evidence capable of determining who can use it competently and effectively, and whether that capability is associated with meaningful individual outcomes. The principal research challenge is thus one of **measurement integration**: connecting dimensions that are currently measured separately without collapsing distinctions that are essential to their interpretation.

The six-stage architecture should therefore be read as an organising measurement framework rather than as a demonstrated causal chain.

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Supplementary Materials / Reproducibility Appendix

A. Reproducibility Checklist

Dataset versions: FITPED AI Literacy Questionnaire data, Figshare Version 2, DOI 10.6084/m9.figshare.29488523.v2; accompanying questionnaire PDF; Skalka et al. (2025a) data article.

Official data sources: Eurostat isoc_ai_iaiu; ITU DataHub; World Bank WDI IT.NET.USER.ZS; UNESCO UIS; OECD PIAAC Cycle 2; IEA ICILS 2023; Pew Research Center; Stanford AI Index 2026; ILO 2025; World Bank/Semrush GenAI-adoption evidence.

Retrieval / consultation information: Public web sources used for final reconciliation were checked on 2 October 2026. Dataset observation years are retained as reported in the manuscript; retrieval date is not treated as the observation year.

Cleaning record: 1,205 records; 1,146 unique participants; 59 repeated IDs; 0 exact duplicate rows; 0 raw missing cells; zero-coded non-applicable responses in IM, S, C, and BI treated as missing for scale analysis; raw dataset not overwritten.

Scale transformation: AI-literacy score = mean of valid L1–L5 items, with a minimum 80% valid-item rule. Original five-item structure retained. L4 deletion was evaluated only as a sensitivity statistic.

Analytical-N rule: Effective analytical Ns are determined by the prespecified ≥80% valid-item rule and may differ from the original FITPED data article's reporting convention.

Variable dictionary: Core variables include participant identifier, year, age, gender, country, study programme, AI-course hours, AI-literacy items L1–L5, and the documented FITPED constructs: AI readiness, relevance, career motivation, social goods, intrinsic motivation, AI anxiety, confidence, satisfaction, and behavioural intention.

Analysis record: FITPED_STEP5_EMPIRICAL_ANALYSIS.xlsx and STEP5_CROSS_DATASET_EMPIRICAL_EVIDENCE_MATRIX.xlsx were the locked analysis/evidence records used for manuscript reconciliation.

Analysis code: No executable analysis-code artifact was present in the supplied manuscript. No code was reconstructed or fabricated during final assembly.

Table-generation record: Original table objects were consolidated into the final seven-table architecture without changing locked numerical values. Group, year, matched-subset, country, capability-context, and audit information not represented as final tables remains in the manuscript prose or is documented in this appendix.

Figure-generation record: Four publication-quality figure objects were generated from the locked manuscript specifications and inserted into the final DOCX. Figure 1 is conceptual/measurement only; Figures 2–4 preserve the documented evidence architecture and locked FITPED values. No respondent-level dataset pooling, unsupported estimate, or causal pathway was introduced.

Respondent-level integration: No FITPED observations were merged with Eurostat, ITU, WDI, UNESCO, PIAAC, ICILS, Pew, Stanford, ILO, or World Bank/Semrush observations.

Composite index: No composite Access–Capability–Effective Use–Benefit index was constructed.

Causal inference: No causal pathway was estimated. The six-stage architecture is conceptual/measurement-based and not an empirically demonstrated causal sequence.

B. Figure 1 Approved Specification

Figure 1. Digital Divide → AI Access Divide → AI Literacy Divide → Capability Divide → Effective-Use Divide → Benefit Divide.

Purpose: provide an integrated measurement architecture for organising evidence across access, perceived literacy, demonstrated capability, effective use, and realised benefit.

Interpretive boundary: conceptual/measurement architecture only; not an empirically demonstrated causal pathway. Figure 1 is now inserted in the main manuscript.

C. Source DOI / URL Register

FITPED dataset: https://doi.org/10.6084/m9.figshare.29488523.v2

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ILO GenAI exposure analysis: https://www.ilo.org/resource/news/one-four-jobs-risk-being-transformed-genai-new-ilo–nask-global-index-shows

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